Dynamics of meaning: Towards the Evaluation of Diachronic Semantic Change in Sinhala

Anonymous ACL submission

Abstract

Tracking semantic change in low-resource languages across extensive historical timelines presents significant challenges due to data scarcity and the limitations of static embedding alignments. This study investigates the diachronic evolution of the Sinhala language from the 13th to the 20th century using a multi-stage computational framework. We first align century-specific `Word2Vec` and `FastText` embeddings using Similarity Matrix Based Alignment (SMA) and Orthogonal Procrustes (OP) techniques, finding that OP alignment provides more stable neighbourhood tracking for identifying temporal similarity dips. To move beyond aggregate measures, we introduce a Bidirectional Semantic Impact Pruning approach using contextualised embeddings from `XLM-R`. By applying Leave-One-Out (LOO) diagnostics, we attempt to isolate influential sentences to distinguish between systemic semantic shifts and transient polysemic expansion. Our results show that semantic drift in `XLM-R` is not evenly distributed across all usages. Instead, a significant part of the change is driven by a smaller set of high-impact contextual instances, rather than gradual and uniform change across all occurrences. This work provides a preliminary framework for diachronic analysis in low-resource contexts, highlighting the trade-offs between model sensitivity and data availability.

1 Introduction

Language is an evolving system that is continuously reshaped by its speakers. Various factors drive this evolution, including phonetic convenience and sociocultural changes (Keidar et al., 2022). In all languages, words naturally exhibit a range of meanings, and the prevalence of these meanings can vary based on the genre and register of discourse, as well as historical context (Frermann and Lapata, 2016). A well-known example of this is the transformation of the word "gay," which has shifted in meaning from "cheerful" to "homosexual" over the years (Beinborn and Choenni, 2020).

The Sinhala language is an Indo-European language with a rich and diverse literary heritage that has developed over several millennia. Its origins can be traced back to between the 3rd and 2nd centuries BCE. The Sinhala language, which is the primary language of the Sinhalese people who constitute the largest ethnic group in the island nation of Sri Lanka, is recognised as the first language (L1) for about 16 million individuals (de Silva, 2025). Sinhala is classified as a lower-resourced language (Category 02) according to the criteria presented by Ranathunga and de Silva (2022). This language has undergone significant evolution and transformation throughout its history, resulting in the modern Sinhala we engage with today.

In this study, we perform a diachronic semantic change analysis using the largest Sinhala diachronic corpus available to date, known as `SiDiaC-v.2.0`. First, we enhance this corpus to create a lemmatised and POS-tagged version, referred to as `SDC`. We then develop diachronic embeddings utilising both static methods and contextualised methods. This is accompanied by various semantic mapping approaches, taking into account the extreme data scarcity encountered per century in the corpus. Finally, we conclude with an evaluation of semantic change across centuries, based on embedding similarity, while considering the best-performing model-to-mapping combinations.

2 Related Works

Research into diachronic semantic change has evolved from establishing fundamental statistical laws to developing sophisticated unsupervised detection frameworks. However, there remains a significant gap in these studies concerning data-scarce settings. Specifically, there is a noticeable correla-

tion between corpus size and the frequency of diachronic semantic change evaluation studies, indicating that studies involving these smaller datasets receive little attention as discussed in Appendix A.

2.1 Semantic Drift Evaluation Frameworks

Foundational work by Hamilton et al. (2016b) proposed the "*law of conformity*," which states that frequently used words change slowly, and the "*law of innovation*," which says that polysemous words change rapidly. They also demonstrated that global shifts in word embeddings capture regular linguistic drift, while changes in local neighbourhoods are more effective at detecting irregular cultural shifts (Hamilton et al., 2016a).

To overcome the limitations of analysing historical time slices in isolation, Frermann and Lapata (2016) developed SCAN, a dynamic Bayesian model that monitors the prevalence of word senses over continuous time intervals. By utilising Gaussian Markov Random Fields, the model ensures that adjacent temporal representations are co-dependent, framing semantic shifts as smooth, gradual processes that inherently enhance sparse historical data. Evaluated on a text corpus spanning from 1700 to 2010, this temporal interdependency allowed SCAN to outperform temporally isolated baselines, achieving a Spearman correlation of 0.377 in detecting changes in word meaning, compared to the baseline's 0.255. Additionally, the model achieved highly competitive results on the SemEval-2015 temporal classification benchmarks, reaching an accuracy of 0.748 in 12-year interval predictions.

Rudolph and Blei (2017) introduced *Dynamic Bernoulli Embeddings* (DBE) to address the limitations of static word representations, which use a Gaussian random walk prior to model semantic evolution across continuous time slices. By treating embedding vectors as sequential latent variables, this architecture effectively accommodates and smooths out temporally sparse historical data. The dynamic model was evaluated across three diverse corpora, including a 151-year dataset of U.S. Senate speeches that comprises 13.7 million words. It consistently outperformed both static and independently trained time-binned baselines in held-out likelihood predictions. For instance, it achieved a better predictive score of -2.4 compared to the -2.454 score of the time-binned baseline on an Arxiv[1] dataset. Additionally, by quantifying absolute semantic drift, the model effectively identifies terms that have undergone significant historical changes. Notably, it mathematically identified "Iraq" as the word that experienced the most substantial shift in the Senate corpus, with a peak drift score of 3.09.

For lexical semantics tasks such as SemEval-2020 (Schlechtweg et al., 2020), systems like SenseCluster (Cuba Gyllensten et al., 2020) utilised multilingual contextualised embeddings (XLM-R) combined with K-Means++ clustering. This approach treats clusters as proxies for senses, enabling the identification of both binary and graded changes in meaning. To address the high computational costs associated with deep learning methods and the limitations of analysing only adjacent time periods, recent research has introduced a framework that uses diachronic word similarity matrices derived from aligned PPMI-SVD embeddings (Kiyama et al., 2025). This framework enables a detailed analysis of continuous semantic shifts and facilitates unsupervised clustering of words with similar temporal trajectories.

2.2 Low-Resource Semantic Drift Evaluation

Addressing the need for evaluating lexical semantic change in small texts, Ehrenworth and Keith (2023) explores unsupervised detection models on corpora of about 150,000 tokens. They downsampled the SemEval-2020 Task 1[2] datasets while maintaining gold-standard annotations and tested three modeling approaches: static SGNS embeddings with orthogonal Procrustes using cosine distance (Pražák et al., 2020), static SGNS with Euclidean distance (Pömsl and Lyapin, 2020), and a fine-tuned BERT model with a temporal self-attention mechanism (Rosin and Radinsky, 2022). The results showed a significant 67% average performance drop in these models in data-scarce environments, though the BERT model remained the most stable. Despite low intrinsic metrics, the researchers highlight the practical utility of these models through a task called "literary intertextual semantic change detection." A case study comparing Frantz Fanon[3] and Saidiya Hartman[4] revealed that the contextual model effectively iden-

[1] https://arxiv.org/

[2] https://www.ims.uni-stuttgart.de/en/research/resources/corpora/sem-eval-ulscd/

[3] https://en.wikipedia.org/wiki/Frantz_Fanon

[4] https://en.wikipedia.org/wiki/Saidiya_Hartman

tifies nuanced semantic shifts in terms like "violence" and "political."

To address noise and overfitting in temporally sparse data, Montariol and Allauzen (2019) evaluated three dynamic embedding models (Incremental Skip-Gram (ISG) (Mikolov et al., 2013; Kim et al., 2014), Dynamic Filtering of Skip-Gram (DSG) (Bamler and Mandt, 2017), and Dynamic Bernoulli Embeddings (DBE) (Rudolph and Blei, 2017, 2018)) on artificially constrained subsets of the New York Times Corpus (Sandhaus, 2008), simulating data scarcity down to 20,000 words (1% subset) per year. Their analysis revealed that a "backward external" initialisation, utilising modern, large-scale pre-trained vectors like Wikipedia (Li et al., 2017) and updating sequentially backward in time, significantly improves performance for sparse historical texts. While the probabilistic dynamic models (DSG and DBE) were far more robust than ISG at capturing directed semantic drifts under extreme scarcity, they initially struggled to isolate extreme shifts. To resolve this, they introduced a *Hardshrink* regularisation term that successfully penalises low-magnitude noise, enabling the models to accurately discriminate valid semantic changes even in severely data-limited environments.

3 Methodology

We began by performing lemmatisation and POS tagging on a subset of SiDiaC-v.2.0, specifically focusing on the documents that include written date annotations. For the evaluation of semantic change, we applied two different approaches to this same subset of SiDiaC-v.2.0.

3.1 Creation of SDC

Our diachronic analysis utilises the SiDiaC-v.2.0[5] corpus (Jayatilleke et al., 2026), which is the successor to SiDiaC-v.1.0[6] (Jayatilleke and de Silva, 2025a). SiDiaC-v.2.0 contains a total of 241,491 words from 185 documents. It is important to note that a book may have been written centuries earlier, while its printed version was released much later (Jayatilleke and de Silva, 2025a). Therefore, for this study, we will focus on a subset of the corpus that includes the written dates, comprising 59 books with a total of 67,005 words, spanning from the 5th century to the 20th century. Further information about these corpora can be found in Appendix B.

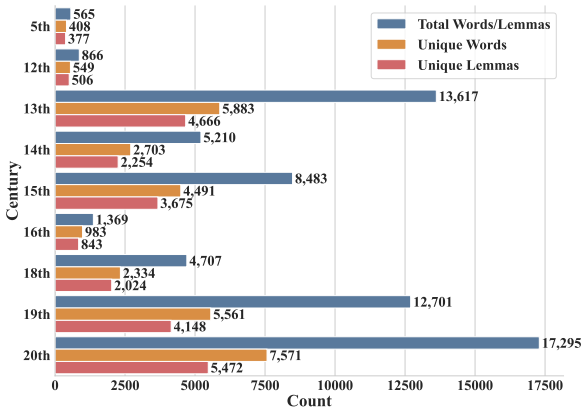


Figure 1: Word Token Counts by Century from the SDC Corpus.

In this study, we enhance the filtered version of the SiDiaC-v.2.0 corpus by adding two new features. Specifically, we provide lemmas for each word based on morphological analysis and include POS tags for every word. As a result, we present the improved version of the corpus, an enhanced *Sinhala Diachronic Corpus* (SDC)[7].

To create this improved SDC, we utilised SinMorphy-v.1.2[8] (Kumarasinghe et al., 2021) for lemmatisation. For POS tagging, we developed a composite pipeline that combines SinMorphy-v.1.2 with an improved version of the *TnT POS Tagger for Sinhala*[9] (Fernando and Ranathunga, 2018) and the Sinling[10] library. A detailed explanation of this enhancement and the new features is provided in Appendix C.

3.2 Diachronic Embeddings

This section discusses three vectorisation techniques used to construct diachronic word embeddings: Word2Vec, FastText, and transformer-based contextualised embedding models. The 5th, 12th, and 16th centuries were excluded during the semantic change study due to insufficient word counts, as illustrated in Figure 1. While the 16th century falls within the 13th–20th century range and was initially considered, its small corpus size reduced the number of consistent words shared across all centuries to 80; removing it raised this to 210. The final analysis, therefore, spans six

[5] https://github.com/NeviduJ/SiDiaC-v.2.0
[6] https://github.com/NeviduJ/SiDiaC
[7] https://anonymous.4open.science/r/Sinhala-Semantic-Change-1FDC
[8] http://nlp-tools.uom.lk/sin-morphy/
[9] https://github.com/nlpcuom/Sinhala-POS-Tagger
[10] https://github.com/ysenarath/sinling

century-specific corpora: the 13th, 14th, 15th, 18th, 19th, and 20th centuries.

3.2.1 Word2Vec Embeddings

Word2Vec embeddings were generated by training a separate model on each century's corpus using a *Skip-gram* architecture (Mikolov et al., 2013), which is well-suited to small and sparse corpora as it learns useful representations even for infrequent terms (Hamilton et al., 2016b). Each model used a vector size of 50, a context window of 5, and Negative Sampling (negative=15) to handle low-frequency vocabulary. A subsampling threshold of 10^{-5} was applied to prevent high-frequency tokens from dominating training (Herbelot and Baroni, 2017), with the learning rate decaying linearly from 5×10^{-2} to 7×10^{-4} over 200 epochs. This produces distinct word spaces per century, enabling diachronic comparison of word representations across contiguous historical time intervals.

3.2.2 FastText Embeddings

A pre-trained FastText model (Bojanowski et al., 2017) trained on the *Sinhala Common Crawl* corpus (cc.si.300.bin) was used as a base model, providing broad lexical coverage grounded in modern Sinhala. The vocabulary of the base model was then extended with any century-specific terms via incremental vocabulary building, and the models were fine-tuned on each century sub-sets for 30 epochs with a linearly decaying learning rate ($1 \times 10^{-2} \rightarrow 1 \times 10^{-4}$). Following training, vocabulary pruning was applied to restrict the model's word vector space to words specifically in that century's corpus, discarding all other words from the pre-trained FastText vocabulary. This procedure yields a set of temporally aligned yet independently fine-tuned embedding spaces.

3.2.3 Contextualised Embeddings

We have examined five Small Language Models (SLMs), with only one being an encoder-only model (XLM-R) and the remaining four being decoder-only transformers. The decoder-only models include three variants of Llama with 8 billion parameters each (base, instruct, and SinLlama (Aravinda et al., 2025)) and Gemma with 9 billion parameters. We conducted Continual Pre-Training (CPT) on all models except for Gemma, due to its low performance on perplexity and Bits-Per-Byte (BPB) scores on the non-lemmatised SDC corpus. The continual pre-training utilised a dataset comprising 90% training data and 10% validation data, based on the full SiDiaC-v.2.0 corpus. Additional information on continual pre-training can be found in Appendix D.1.

Model	Perplexity	Bits-Per-Byte
Llama-3.1-8B	**2.86**	1.04
Gemma-2-9b	25.17	1.49
SinLlama	186.67	0.92
Llama-3.1-8B-it	3.42	1.22
XLM-R	-	**0.80**
Llama-3.1-8B-FT	623.76	1.22
SinLlama-FT	203103.64	1.96
XLM-R-FT	-	1.10

Table 1: Evaluation of SLMs on SiDiaC-v.2.0 based on Perplexity and Bits-Per-Byte Metrics.

Contextual word embeddings were extracted using the best-performing pretrained transformer model, the base XLM-R (Conneau et al., 2020). This multilingual encoder operates in full precision and was evaluated based on the BPB scores presented in Table 1. It is important to note that XLM-R does not produce traditional perplexity scores. This is because its bidirectional masked language modelling objective computes loss only for isolated masked tokens, rather than modelling the joint probability of an entire sequence of text. This information highlights that, despite the emergence of larger models, an older encoder-only model can still better grasp the nuances of low-resource languages. A detailed analysis of the perplexity and BPB scores can be found in Appendix D.2.

The model produces word-level embeddings from the final hidden layer (768 dimensions). Since XLM-R employs a subword tokeniser, sub-token vectors for each word were aggregated via mean pooling following the methodology by Cuba Gyllensten et al. (2020). To handle long sentences, a sliding window of 510 tokens with a step of 300 tokens was applied, with overlapping regions averaged to reduce boundary effects. This process was applied independently per century for both models, yielding century-specific contextual word representations that capture semantic variation across distinct historical time intervals. It is important to note that, unlike Word2Vec and FastText approaches, where we had trained models for each century, we leverage the frozen pretrained representations following Martinc et al. (2020). We hypothesise that the high-dimensional embedding space contains inherent low-dimensional manifolds representing century-specific semantic shifts, thereby bypassing the need for period-specific retraining with small datasets, which could lead to overfitting and align-

ment issues.

3.3 Computational Semantic Mapping

This section outlines the semantic mapping methodology used to compare word representations across centuries. We employ two complementary alignment strategies for the `Word2Vec` and `FastText` embeddings. Such alignment is necessary because these models use randomised initialisation and stochastic training, resulting in vector spaces that are arbitrarily rotated relative to one another (Hamilton et al., 2016b), a problem known as non-identifiability (Carrington et al., 2019). A summarised illustration of the alignment mechanisms is shown in Figure 4 in Appendix E.

3.3.1 Similarity Matrix Based Alignment

Despite the fact that vector spaces can be "arbitrarily rotated," the similarity between two words within one embedding space can be directly compared to that in another space (Hamilton et al., 2016b). This means that even if the axes are rotated, the distance between two vectors remains consistent. Based on this principle, we decided to create a similarity matrix for each word in one space, which can then be compared to another matrix of the same word from a different space.

To account for morphological variation in Sinhala, we first construct a centroid-based lemma representation: for each lemma L with surface forms $\{w_1, w_2, \ldots, w_n\}$, its vector is computed as the arithmetic mean of its valid constituent word vectors, collapsing morphological variance into a single stable point. For cross-century comparison, we adopt a relational similarity approach using 261 common lemmas shared across all century spaces as a common reference frame. Each word is represented by a *similarity profile*, a vector of cosine distances to every anchor lemma, and the *Pearson* correlation between the profiles of the same word in two centuries is used as the measure of semantic similarity, bypassing the need for explicit space alignment.

3.3.2 Orthogonal Procrustes Based Alignment

The concept of unitary invariance in word embeddings suggests that two embeddings are essentially identical if one can be derived from the other through a unitary operation, such as a rotation (Yin and Shen, 2018). The relative structure and relationships between words remain unchanged if all vectors are transformed by an orthogonal matrix. This preservation of structural integrity ensures that the embeddings retain their usefulness in tasks that rely on cosine similarity, even after these transformations (Carrington et al., 2019). Based on this property, we align century-specific word spaces using two variants of OP analysis: a conventional statistical formulation and a neural network-based formulation.

For rotational alignment, each anchor must be a single, fixed vector in the embedding space, making centroid-based lemma representations unsuitable. The rotation matrix Q relies on a one-to-one correspondence between vectors, so we use the 210 consistent surface-form words as anchors across all century models. This alignment mechanism was conducted based on two approaches:

Analytical Procrustes Alignment: For each century space, an anchor matrix is constructed from the vectors of these 210 words and subsequently L2-normalised. The optimal rotation matrix $Q = UV^\top$ is derived via Singular Value Decomposition (SVD) of the cross-covariance between the target and reference anchor matrices, rotating the target space to minimise $\|W_{\text{target}}Q - W_{\text{ref}}\|_F$ while preserving all internal geometric relationships. The 13th-century space serves as the global reference anchor, and all other century spaces are permanently rotated in place to this coordinate system, allowing direct word-level comparison across centuries.

Neural Procrustes Alignment: As a complementary approach, we frame the OP problem as a supervised learning task using a bias-free, activation-free linear layer $\mathbf{W} \in \mathbb{R}^{d\times d}$, initialised as the identity matrix and trained to map L2-normalised anchor embeddings from a target century into the coordinate system of a reference century. The training objective combines (Mean Squared Error) MSE for alignment with a squared *Frobenius-norm* orthogonality penalty:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda\|\mathbf{W}\mathbf{W}^\top - \mathbf{I}\|_F^2 \qquad (1)$$

where $\lambda = 0.001$ provides the necessary flexibility for the optimiser to minimise alignment error across both `Word2Vec` and subword-informed `FastText` architectures without inducing geometric distortion, although higher values of λ theoretically enforce a more rigid rotation. The model is optimised with Adam at a learning rate of 0.001, chosen to ensure stable convergence of the orthogonality constraint. Training runs for up to 2000

epochs with early stopping (patience of 20 epochs, minimum 100 epochs). A separate $\mathbf{W}$ is learned for each target century, with the reference set to the 13th century, yielding five transformation matrices across six centuries.

Model	Approach	Avg. Procrustes Disparity	Avg. Rotational Distance
Word2Vec	Analytical OP	174.16	9.93
	Neural OP	157.16	9.16
FastText	Analytical OP	57.57	22.82
	Neural OP	57.06	19.68

Table 2: Evaluation of Orthogonal Procrustes Alignment Performance.

We evaluate embedding alignment using *Procrustes Disparity* (Rochkoulets et al., 2026), which measures the residual error $\|W_{target}Q - W_{ref}\|_F^2$ after orthogonal transformation, and *Rotational Distance*, defined as $\|Q - I\|_F$ to quantify the magnitude of the rotation Q. As shown in Table 2, the *Neural OP* approach for FastText significantly outperforms the *Analytical OP* used for Word2Vec. This neural-based optimisation achieves better manifold preservation, yielding markedly lower error margins for both disparity and distance scored compared to the higher absolute residuals of the analytical baseline. While having more data is essential for improving embedding stability, the detailed analysis of the target-to-reference century alignment provided in Appendix E.1 indicates that Word2Vec worsens with increased data exposure, unlike the fine-tuned FastText models. Therefore, we conclude that FastText should be prioritised in the statistical neighbourhood analysis discussed in subsection 4.1.

4 Evaluation of Semantic Change

4.1 Statistical Neighbourhood Analysis

To systematically isolate and qualitatively evaluate lexical items that have undergone semantic change in both the SMA (lemma-level) and the two OP Alignment (word-level) approaches, we implemented an automated reporting mechanism. Due to variations in corpus sizes and differences in the inherent alignment quality across temporal periods, applying a static cosine similarity threshold across all periods was suboptimal. Instead, we established a dynamic, pair-specific approach to define cutoffs for semantic drift.

For any given pair of time periods, we calculate the mean (μ) and standard deviation (σ) of the cosine similarities for all intersecting vocabulary items between the two aligned vector spaces. A word or a lemma is then classified as having undergone significant semantic change if its similarity score falls below a dynamically defined threshold, calculated as $\mu - (k \times \sigma)$, where k is a scaling factor that controls the stringency of the cutoff. In our analyses, we empirically set k to 0.8 to achieve a balance between capturing genuine language changes and avoiding meaningless background noise. Setting k lower risks including stable words that may have shifted solely due to alignment limitations, since even historically consistent keys achieve a mean baseline similarity of 52.92% for Word2Vec and 81.04% for FastText under Neural OP, as shown in Table 3. Conversely, setting k higher makes the filter too strict, resulting in the exclusion of valid drifting words. This approach effectively accounts for the variance inherent in each specific pair of models.

	SMA	Analytical OP	Neural OP
Word2Vec	30.23%	61.31%	52.92%
FastText	55.69%	82.11%	81.04%

Table 3: Mean similarity of consistent keys across 15 temporal alignments using Word2Vec and FastText for SMA and OP approaches.

Following the quantitative identification of drifting candidates, we perform a qualitative contextualisation step. For each identified word, we retrieve its top 5 nearest neighbours in both the source and target semantic spaces. By comparing these lists of nearest neighbours side by side, we can manually interpret the specific nature of the semantic shift, observing how a lemma's or a word's contextual usage and associative meanings have transitioned over time. This methodology is applied uniformly across both alignment models, ensuring consistent qualitative insights regardless of the underlying alignment technique.

We evaluate six combinations using Word2Vec and FastText across three alignment strategies: SMA and two OP variants. Following Gulordava and Baroni (2011), we expect high mean similarity in temporal semantic spaces, as most of the lexicon remains stable while semantic change affects only a small vocabulary subset. Our results indicate high similarity in FastText methods (see Table 3). However, this could be due to fine-tuning on pretrained models to compensate for the small corpus. Thus, we must consider both Word2Vec and FastText, given their unique strengths and weaknesses.

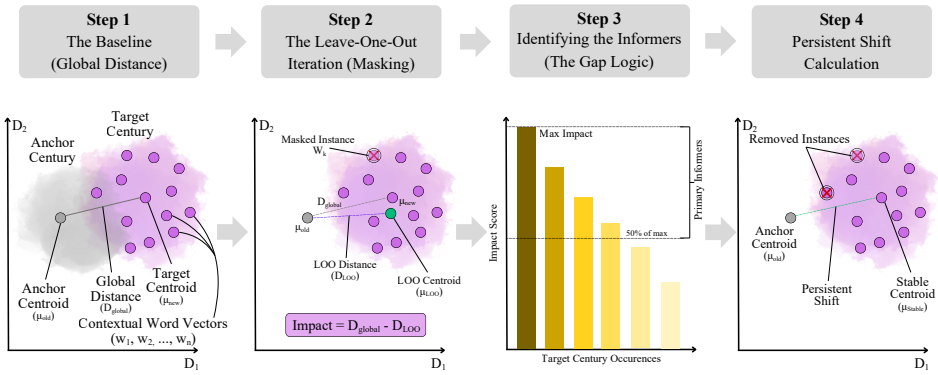


Figure 2: Leave-One-Out (LOO) analysis for identifying contextual semantic divergence.
Step 1: The baseline global distance is established. **Step 2:** Individual contextual word vectors (w_k) are iteratively masked to measure their impact on centroid displacement. **Step 3:** Textual evidence of semantic divergence is extracted as "primary informers" based on this impact. **Step 4:** Informers are permanently removed to isolate the word's persistent shift in core meaning.

In contrast, SMA appears less robust based on the mean similarity statistics in Table 3 compared to OP techniques. This is likely because SMA considers all consistent lemmas when calculating relational similarity. This global approach includes both stable anchor lemmas and those that have undergone semantic change, meaning the resulting relational similarity shifts are distorted by those localised drifts.

We analysed a total of 15 century-aligned semantic spaces, each containing 261 consistent lemmas and 210 consistent words. This resulted in 2,116 instances of similarity for all combinations of models and alignment approaches for each century-to-century alignment. During our analysis, we found that 229 instances of these lemmas were captured using the `Word2Vec` model and 59 using `FastText` through the SMA approach. Additionally, we identified 635 word instances from `Word2Vec` and 271 from `FastText` using the analytical procrustes approach. Furthermore, the neural procrustes approach revealed 648 word instances from `Word2Vec` and 274 from `FastText`.

The high frequencies of temporal similarity dips observed in the `Word2Vec` embeddings, along with the similarly distributed frequencies across the 15 aligned semantic spaces (as shown in Table 10), again indicate that `Word2Vec` embeddings were less stable compared to `FastText` embeddings.

For example, using `FastText` and neural procrustes alignment, we can observe that the word 'රැස් \ ræs' in the 14th century appears in a semantic environment associated with gathering, movement, and central positioning. Its neighbors include causing to gather (රැස්කරවා \ ræskʌrʌvɑː), receiving or accepting (පිළිගෙණ \ pɪlɪgɛnʌ), in the middle (මධ්‍යයෙහි \ mʌðʰyʌjɛhɪ), and with speed (වේගයෙන් \ veːgʌjɛn). In this context, 'රැස් \ ræs' is linked with the idea of collecting or assembling, often in physical or spatial terms. By the 15th century, the lemma shifts into a more symbolic and doctrinal semantic field. Its neighbors include destruction or end (විනාස \ vɪnɑːsʌ), life span (ආයු \ ɑːjʊ), secret or hidden (රහස්‍ය \ rʌhʌsyʌ), gem or precious object (රත්න \ rʌθnʌ), and the Buddha's radiance (බුදුරැස් \ bʊðʊræs). In this period, 'රැස් \ ræs' is also associated with spiritual and symbolic meanings, especially light, brilliance, and sacred radiance in Buddhist contexts. Further examples from all six model-to-mapping combinations are discussed in Appendix F.1.

4.2 Bidirectional Semantic Impact Scoring

By using created `XLM-R` contextual embeddings, filtered to the six valid centuries between 13th–20th as discussed in subsection 3.2.3, lemma-level analysis was enabled by constructing a sentence-aligned lemma lookup cross-referencing the lemma and word token versions of the corpus: for each occurrence of a word, identified by its sentence and position within each century, we mapped the corresponding lemma to its contextual word vectors and the original sentence texts. This process created a flat record for each lemma occurrence that includes its century, surface-form words, vectors, and original sentence contexts.

From this, a lemma map was constructed per century: for each lemma, all contextual vectors of its surface form words across all sentences were collected, preserving the full distributional varia-

tion of each lemma within a century. The lemma centroid, computed as the arithmetic mean of all instance vectors, serves as the aggregate representation for cross-century comparison. Critically, the individual instance vectors and their source texts are retained alongside the centroid.

To interpret the specific linguistic context driving the numerical change observed between chronological centroids generated from XLM-R embeddings, we implemented a diagnostic Leave-One-Out (LOO) analysis. The baseline for semantic change is established as the global cosine distance between the anchor century's centroid and the target century's centroid (D_{global}). To isolate the impact of individual textual usages, the algorithm iterates through every vectorised instance of the lemma in the target century. In each iteration, a single sentence instance is provisionally excluded from the target pool, a new centroid (D_{LOO}) is calculated, and the cosine distance to the anchor centroid is re-evaluated.

By subtracting D_{LOO} from the baseline global distance (D_{global}), we quantify the semantic impact of the excluded sentence. A positive impact score indicates that the presence of the sentence actively pulled the target centroid away from its historical anchor meaning. To capture related senses of semantic change, the algorithm identifies the sentence with the maximum displacement impact and flags it, alongside any other instances yielding an impact score within 50% of this maximum, as primary "informers". The 50% threshold was chosen as a mid-level salience criterion to balance specificity and coverage when identifying important contextual usages. Higher thresholds often selected only one dominant sentence, while lower thresholds included many weakly influential instances, which reduced interpretability. Therefore, the 50% cutoff keeps instances whose impact is still close to the strongest effect, capturing the main contributors to semantic change without adding noise. In a forward temporal comparison (lower to upper century), these "informers" represent novel semantic innovations; in a reverse comparison (upper to lower century), they highlight obsolete contextual usages that eventually faded from the lexicon.

Finally, out of the total instance count for the target lemma, these isolated informer sentences are permanently masked from the target century representations to recalculate a revised distance metric, termed the "persistent shift". The mathematical difference between the original global distance and the persistent shift determines exactly what percentage (Drift Reduction) of the observed semantic drift is attributable to specific documented contextual innovations. A summary of the complete mechanism is illustrated in Figure 2. This LOO methodology, applied uniformly across XLM-R embeddings over all 30 ordered permutations of the six centuries, provides both quantitative measurement and immediate qualitative textual evidence of semantic change.

The results show a low but consistent level of semantic shift in the XLM-R representations. The average global centroid distance is 0.0029, which decreases to 0.0024 after removing high-impact instances, resulting in a drift reduction of 18%. This indicates that a measurable portion of the observed shift is due to a subset of high-influence contextual usages. Out of 108,334 instances, a total of 22,727 are identified as having drifted, revealing that semantic change is unevenly distributed across occurrences. Overall, these findings support the idea that diachronic semantic drift in this model is driven by selective contextual innovations rather than uniform shifts across all instances. A detailed analysis of these metrics, based on all 30 ordered permutations of the six centuries, along with examples of semantic change, is discussed in Appendix F.2.

5 Conclusion

This study established a computational framework for diachronic semantic analysis using three distinct embedding methodologies. We successfully aligned century-specific Word2Vec and FastText models through SMA and OP alignment techniques. Our evaluation of these static embeddings via statistical neighbourhood analysis demonstrated the superior stability of OP alignment for identifying temporal "similarity dips" in the Sinhala lexicon.

Furthermore, we extended the analysis beyond aggregate centroid-based measures by introducing *Bidirectional Semantic Impact Pruning*. Utilising contextualised embeddings from XLM-R, this approach leveraged LOO diagnostics to isolate and identify sentences responsible for semantic drift. In contrast to global averaging, this method attempts to distinguish between systemic shifts and polysemic expansion by highlighting specific influential instances. While limited by available data density, the approach offers a preliminary framework for exploring the semantic evolution of low-resource languages like Sinhala across historical periods.

Limitations

The scope and outcomes of this study on Sinhala diachronic semantics were influenced by several constraints.

Data Scarcity: The lack of extensive historical corpora impacted the strength of century-wise `Word2Vec` models, leading to possible problems such as unreliable *Procrustes* alignment compared to the fine-tuned `FastText` models. Additionally, the transformer models did not improve the perplexity and BPB scores during continual pretraining, possibly due to data scarcity affecting the embedding strength of expanded vocabulary words.

Lemmatisation: `SinMorphy` may not have been trained on historical Sinhala. The absence of robust morphological analyzers highlights the low-resource status of Sinhala (de Silva, 2025).

POS Tagging: The absence of high-accuracy POS taggers for historical Sinhala (de Silva, 2025) necessitated a hybrid tagging system. This system likely carries a modern linguistic bias, as its dependencies are rooted in contemporary Sinhala grammar, potentially affecting the accuracy of morphological labelling within the medieval and early modern sections of the corpus.

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A Corpora vs Semantic Drift Studies

In this study, we analysed the corpora used for diachronic semantic change research based on the list identified by Jayatilleke et al. (2026). A total of 81 corpora were listed, including the proposed `SiDiaC-v.2.0`. The phenomena examined in the works surveyed by Tahmasebi et al. (2018) regarding computational approaches to lexical semantic change detection are tagged under the following headings: lexical (semantic) change, grammaticalisation, and lexical replacement. Based on these tags, we identified 50 corpora that have been used at least once in a semantic change study.

Based on our findings, it is important to highlight that `SiDiaC-v.2.0` is one of three corpora that have undergone studies on semantic change for any language classified as language class 3 or below, according to the language resource taxonomy by Ranathunga and de Silva (2022) as shown in Figure 3. Among these three corpora, `SiDiaC-v.2.0` is unique in its general study of lexical semantic change, as it does not focus on specific instances of grammaticalisation like the other two corpora. Jayatilleke et al. (2026) conducted this semantic change evaluation on `SiDiaC-v.2.0` using a statistical Bag of Words (BoW) analysis during the proposal of the corpus. This emphasises that low-resource languages are often overlooked in the field of diachronic semantic change, highlighting a significant gap in this area of research. A comprehensive summary of the surveyed corpora by Jayatilleke et al. (2026), along with their status regarding lexical semantic change studies and several example research articles, can be found in Table 4.

B `SiDiaC`: Sinhala Diachronic Corpus

B.1 `SiDiaC-v.1.0`

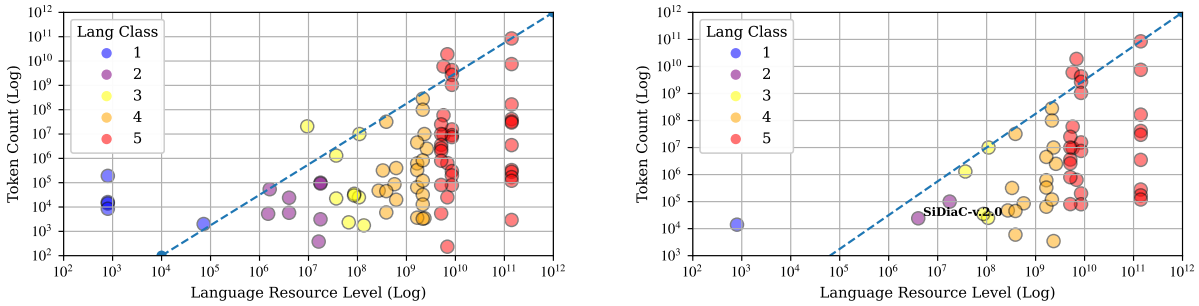
`SiDiaC-v.1.0` (Jayatilleke and de Silva, 2025a) is the first comprehensive diachronic corpus for the Sinhala language, consisting of approximately 58,000 word tokens extracted from 46 literary works that span from the 5th to the 20th century CE. The dataset was created by digitising texts acquired from the National Library of Sri Lanka using the `Google Document AI` (Jayatilleke and de Silva, 2025c) OCR engine. This engine performed crucial functions such as text modernisation and morpheme segmentation to effectively manage historical orthography.

A key feature of `SiDiaC-v.1.0` is its annotation based on the written date, which was determined through the lifespans of authors and secondary historical sources like the *Sinhala Sahithya Wanshaya* (Sannasgala, 2015), rather than relying on the printed issue date. This approach ensures a precise timeline for historical analysis, despite the challenges associated with dating undated manuscripts.

The corpus is organised with a two-level genre classification system: a primary layer that distinguishes between Fiction and Non-Fiction, and a secondary layer that categorises texts into Religious, History, Poetry, Language, and Medical genres. The content reflects Sri Lanka's literary history, showcasing a predominance of religious and poetic texts due to the influence of Theravada Buddhism and courtly patronage. It also retains code-mixed passages in Pali and Sanskrit to preserve the original context.

B.2 `SiDiaC-v.2.0`

`SiDiaC-v.2.0` (Jayatilleke et al., 2026) is the largest comprehensive diachronic corpus for Sinhala to date, significantly expanding upon its predecessor. It includes 244,204 word tokens across



(a) 81 listed diachronic corpora

(b) Corpora that have undergone semantic drift evaluation.

Figure 3: Log token counts of the existing diachronic corpora listed by Jayatilleke et al. (2026) (including `SiDiaC-v.2.0`) against the log of language resource level.

186 literary works. The full corpus spans publication dates from 1800 to 1955 CE; however, a critically important subset of 59 documents, comprising 67,005 words, has been annotated with precise written dates ranging from the 5th to the 20th century CE. This annotation facilitates accurate historical analysis.

Similar to `SiDiaC-v.1.0` (Jayatilleke and de Silva, 2025a), text extraction was performed using the `Google Document AI` (Jayatilleke and de Silva, 2025c) OCR engine. This tool was chosen for its advanced capabilities in text modernisation and morpheme segmentation, which are necessary for processing historical Sinhala orthography. The corpus follows a two-level genre classification system. It first distinguishes between Fiction and Non-Fiction and then further categorises texts into specific domains such as Religious, History, Poetry, Language, and Medical. A frequency analysis illustrating the distribution of resources across genres and written time periods, represented in centuries, is shown in Table 5.

To ensure linguistic integrity, the dataset underwent deeper post-processing by native speakers. This process involved removing code-mixed content (including Pali, Sanskrit, and English) and correcting formatting issues. Additionally, special tokens such as `<eos>` for sentence boundaries and `<psi>` to preserve the unique suffix structure of Sinhala poetry were introduced.

C Creation of SDC

C.1 Lemmatisation

Lemmatisation was performed on the filtered `SiDiaC-v.2.0` subset using `SinMorphy-v.1.2`[11] (Kumarasinghe et al., 2021), which is a morphological analyser. Since `SinMorphy` is accessible only through a web interface, we used that platform to perform morphological analysis on our sentences from each century. The automation process operates using a `Selenium`[12] based browser instance to sequentially process tokenised sentences. It incorporates adaptive retry logic to reduce the impact of network latency and potential timeouts during server interaction.

`SinMorphy` (Kumarasinghe et al., 2021) is a rule-based system that relies on a *Finite State Transducer* (`FST`) (Beesley and Karttunen, 2003) to process words independently, without employing deep learning or contextual information. The system is implemented using the `Foma` [13] (Hulden, 2009) compiler for general rules and the `LEXC` file format to define the lexicon and specify morphological continuation classes. Its internal structure comprises four main components: a lexicon categorised by POS, a set of morphemes, lexical tags, and rules that dictate operations such as concatenation, deletion, and gemination. For words not explicitly listed in the vocabulary, the system employs a "guesser" component that inspects the final characters of the unknown word to infer and assign likely POS tags based on known morphological patterns.

For each word, we parse the morphological string to isolate the root lemma, selecting the appropriate analysis when multiple analyses are provided. We select all options if the system is uncertain about the POS tag, which is disambiguated after the final POS tagging discussed in subsection C.2. The system also addresses irregular outputs, particularly for unrecognised terms, which are primarily labelled as "proper noun or invalid" by `SinMorphy`.

[11] http://nlp-tools.uom.lk/sin-morphy/

[12] https://www.selenium.dev/

[13] https://fomafst.github.io/

Dataset	Language		Time Span	Token Count	LSC Status	Example LSC Studies
	Name	Class				
ENGALL (Davies, 2012)	English	5	1800.0 – 1999.0	0.09×10^{13}	✓	Hamilton et al. (2016a)
ENGFIC (Davies, 2012)	English	5	1800.0 – 1999.0	0.07×10^{12}	✓	Hamilton et al. (2016a)
People in the News (Hennig and Wilson, 2020)	English	5	2000.0 – 2019.0	0.02×10^{11}	✓	Hennig and Wilson (2020)
COHA (Davies, 2012)	English	5	1810.0 – 2009.0	0.04×10^{10}	✓	Hamilton et al. (2016b) Giulianelli et al. (2020) Fonteyn et al. (2020) Davies (2012)
EDGeS-English (Bouma et al., 2020)	English	5	1301.0 – 2020.0	0.03×10^{10}	✗	
English Scientific Writing (Steuer et al., 2024)	English	5	1665.0 – 1996.0	0.03×10^{10}	✓	Steuer et al. (2024)
Royal Society Corpus (RSC) (Kermes et al., 2016)	English	5	1665.0 – 1869.0	0.04×10^{9}	✓	Degaetano-Ortlieb and Teich (2016)
ARCHER (Biber et al., 1994)	English	5	1700.0 – 1994.0	0.03×10^{8}	✗	
PPCHE-Modern British English (Kroch et al., 2016)	English	5	1707.0 – 1914.0	0.03×10^{8}	✓	Fonteyn and Hartmann (2016) Fonteyn et al. (2025)
PPCHE- Early Modern English (Kroch et al., 2004)	English	5	1500.0 – 1720.0	0.02×10^{8}	✓	Fonteyn and Hartmann (2016) Fonteyn et al. (2025)
PPCHE-Middle English (Kroch and Taylor, 2000)	English	5	1150.0 – 1500.0	0.01×10^{8}	✓	Rodríguez-Puente (2012) Tissari (2008)
ISWOC-English (Bech and Eide, 2014)	English	5	400.0 – 1010.0	0.03×10^{6}	✗	
GERALL (Schneider and Volk, 1998)	German	5	1800.0 – 1999.0	0.04×10^{12}	✓	Hellrich and Hahn (2017) Hellrich and Hahn (2016)
German Court Decisions (Braun, 2022)	German	5	1970.0 – 2020.0	0.03×10^{12}	✓	Braun (2022)
AMC (Jutta et al., 2013)	German	5	1986.0 – 2012.0	0.01×10^{12}	✓	Marakasova and Neidhardt (2020) Baumann et al. (2019)
DTA (Geyken et al., 2011)	German	5	1600.0 – 1899.0	0.02×10^{10}	✓	Schlechtweg et al. (2018) Schlechtweg et al. (2020)
EDGeS-German (Bouma et al., 2020)	German	5	1301.0 – 2020.0	0.09×10^{9}	✗	
ParlAT (Wissik and Pirker, 2018)	German	5	1945.0 – 2017.0	0.07×10^{9}	✓	Yim et al. (2022)
RIDGES (Odebrecht et al., 2017)	German	5	1478.0 – 1870.0	0.03×10^{8}	✗	
ReM (Klein and Dipper, 2016)	German	5	1050.0 – 1350.0	0.02×10^{8}	✓	Breitbarth (2019)
GerManC (Durrell et al., 2012)	German	5	1650.0 – 1800.0	0.08×10^{7}	✓	Whitt (2016) Whitt (2018) Hartmann (2014)
FREALL (Sagot et al., 2006)	French	5	1800.0 – 1999.0	0.02×10^{13}	✓	Hamilton et al. (2016a)
LLODIA-French (Armaselu et al., 2024a)	French	5	1690.0 – 1918.0	0.06×10^{8}	✓	Armaselu et al. (2024b)
SRCMF (Lavrentiev, 2011)	French	5	800.0 – 1299.0	0.03×10^{7}	✗	
ISWOC-French (Bech and Eide, 2014)	French	5	1225.0 – 1275.0	0.02×10^{5}	✗	
CHIALL (Xue et al., 2005)	Chinese	5	1950.0 – 1999.0	0.06×10^{12}	✓	Hamilton et al. (2016b)
ZhShiftEval (Chen et al., 2022)	Chinese	5	1953.0 – 2003.0	0.06×10^{10}	✓	Periti and Tahmasebi (2024) Goworek and Dubossarsky (2026)
People's Daily (He et al., 2014)	Chinese	5	1947.0 – 1996.0	0.10×10^{9}	✓	Mou et al. (2015)
CORDE (Shuger, 2020)	Spanish	5	1472.0 – 1975.0	0.03×10^{10}	✓	Pascual-López (2023)
Corpus del Español (CdE) (Davies, 2002)	Spanish	5	1200.0 – 1999.0	0.01×10^{10}	✓	Korfhagen (2017) Healey (2017)
CorDECh (Contreras Seitz, 2009)	Spanish	5	1500.0 – 1699.0	0.04×10^{9}	✓	Garrido Sepúlveda and Insausti Muñoz (2025) Sepúlveda et al. (2025)
EZLN (Gribomont, 2023)	Spanish	5	1952.0 – 2023.0	0.03×10^{9}	✓	Gribomont (2025)
IAC-Spanish (Sánchez-Marco et al., 2010)	Spanish	5	1100.0 – 1599.0	0.02×10^{9}	✗	
IMPACT-es (Sánchez-Martínez et al., 2013)	Spanish	5	1482.0 – 1990.0	0.08×10^{8}	✓	Melis et al. (2023)
PS Post Scriptum-Spanish (Vaamonde, 2015)	Spanish	5	1500.0 – 1800.0	0.08×10^{7}	✓	Garcia and García Salido (2019) Simonenko and Carlier
ISWOC-Spanish (Bech and Eide, 2014)	Spanish	5	1221.0 – 1492.0	0.05×10^{6}	✗	
EDGeS-Dutch (Bouma et al., 2020)	Dutch	4	1301.0 – 2020.0	0.03×10^{9}	✓	Coussé et al. (2024)
DiaCORIS (Onelli et al., 2006)	Italian	4	1750.0 – 1945.0	0.01×10^{10}	✓	Viola (2017)
GDLI (Favaro et al., 2022)	Italian	4	1300.0 – 1999.0	0.03×10^{6}	✓	Musi (2016)
Kubhist (Lilljegren, 2018)	Swedish	4	1700.0 – 1999.0	0.03×10^{11}	✓	Schlechtweg et al. (2020)
The Swedish Culturomics Gigaword corpus (Eide et al., 2016)	Swedish	4	1950.0 – 2015.0	0.01×10^{11}	✓	Ahrenberg et al. (2022)
EDGeS-Swedish (Bouma et al., 2020)	Swedish	4	1301.0 – 2020.0	0.08×10^{8}	✗	
FSV (Delsing, 2002)	Swedish	4	1276.0 – 1734.0	0.01×10^{8}	✓	Sundquist (2018)
Menota-Swedish (Haugen et al., 2008)	Swedish	4	1400.0 – 1550.0	0.03×10^{7}	✗	
HaCOSSA (Höder, 2012)	Swedish	4	1375.0 – 1550.0	0.01×10^{7}	✗	
MAþiR (Språkbanken Text, 2024)	Swedish	4	1200.0 – 1299.0	0.03×10^{6}	✗	
CdP (Davies, 2009)	Portuguese	4	1200.0 – 1900.0	0.04×10^{9}	✓	Gerhalter and Koch (2025)
Colonia (Zampieri and Becker, 2013)	Portuguese	4	1500.0 – 1999.0	0.06×10^{8}	✓	Amaral et al. (2023)
TBCHP (Galves et al., 2005)	Portuguese	4	1500.0 – 1899.0	0.03×10^{8}	✓	Amaral and Howe (2012)
PS Post Scriptum-Portuguese (Vaamonde, 2015)	Portuguese	4	1500.0 – 1800.0	0.06×10^{7}	✓	Garcia and García Salido (2019)
ISWOC-Portuguese (Bech and Eide, 2014)	Portuguese	4	1344.0 – 1400.0	0.04×10^{6}	✗	
DIAKORP (Kučera et al., 2015)	Czech	4	1300.0 – 1999.0	0.04×10^{8}	✗	
DIALEKT (Kopřivová et al., 2014)	Czech	4	1960.0 – 1989.0	0.02×10^{7}	✗	
Menota-Norwegian (Haugen et al., 2008)	Norwegian	4	1200.0 – 1350.0	0.09×10^{7}	✓	Vindenes (2017)
RuSemShift (Rodina and Kutuzov, 2020)	Russian	4	1682.0 – 2017.0	0.03×10^{10}	✓	Rodina et al. (2020)
RuShiftEval (Kutuzov and Pivovarova, 2021)	Russian	4	1700.0 – 2016.0	0.04×10^{7}	✓	Lidia and Andrey (2021)
TOROT-Russian (Eckhoff and Berdicevskis, 2015)	Russian	4	1400.0 – 1699.0	0.06×10^{6}	✓	Zanchi et al. (2016)
HGDS (Simon, 2014)	Hungarian	4	900.0 – 1499.0	0.03×10^{8}	✓	Egedi and Simon (2012)
RoDICA (Gîfu and Simionescu, 2016)	Romanian	4	1840.0 – 1991.0	0.05×10^{7}	✓	Truică et al. (2023)
Menota-Danish (Haugen et al., 2008)	Danish	3	1300.0 – 1300.0	0.02×10^{6}	✗	
LLODIA-Hebrew (Armaselu et al., 2024a)	Hebrew	3	1000.0 – 2024.0	0.01×10^{10}	✓	Armaselu et al. (2024a)
PROIEL-Ancient Greek (Haug and Jøhndal, 2008)	Greek	3	0.0 – 999.0	0.03×10^{7}	✓	Lavidas and Haugh (2020)
SLIEKKAS (Gelumbeckaitė et al., 2012)	Lithuanian	3	1500.0 – 1800.0	0.04×10^{7}	✓	Armaselu et al. (2024a)
IMP-sl (Erjavec, 2015)	Slovene	3	1584.0 – 1918.0	0.03×10^{7}	✗	
PROIEL-Armenian (Haug and Jøhndal, 2008)	Armenian	3	400.0 – 450.0	0.02×10^{6}	✗	
LatinISE (McGillivray and Kilgarriff, 2013)	Latin	3	-186.0 – 2000.0	0.01×10^{9}	✓	McGillivray (2021) Schlechtweg et al. (2020)
PROIEL-Latin (Haug and Jøhndal, 2008)	Latin	3	300.0 – 499.0	0.02×10^{7}	✗	
GNC (Gippert and Tandashvili, 2015)	Georgian	3	400.0 – 2015.0	0.02×10^{10}	✗	
IcePaHC (Rögnvaldsson et al., 2012)	Icelandic	2	1100.0 – 2012.0	0.01×10^{8}	✓	Egedi (2013)
Menota-Icelandic (Haugen et al., 2008)	Icelandic	2	1200.0 – 1700.0	0.09×10^{7}	✗	
Greinir skáldskapar (Eyþórsson et al., 2014)	Icelandic	2	900.0 – 1270.0	0.03×10^{6}	✗	
Pre-Standard Irish (Scannell, 2022)	Irish	2	1600.0 – 1936.0	0.04×10^{5}	✗	
SiDiaC-v.2.0 (Jayatilleke et al., 2026)	Sinhala	2	400.0 – 1999.0	0.02×10^{7}	✓	Jayatilleke et al. (2026)
SiDiaC-v.1.0 (Jayatilleke and de Silva, 2025b)	Sinhala	2	400.0 – 1999.0	0.06×10^{6}	✗	
DCS (Hellwig et al., 2020)	Sanskrit	2	-1300.0 – -700.0	0.05×10^{7}	✗	
FarPaHC (Rögnvaldsson et al., 2012)	Faroese	2	1800.0 – 2012.0	0.05×10^{6}	✗	
DACON (O'Neill and Meelen, 2024)	Newar	1	1114.0 – 1899.0	0.02×10^{6}	✗	
DIACU-Old Church Slavonic (Cassese et al., 2025)	Slavonic	1	800.0 – 1799.0	0.02×10^{8}	✗	
TOROT-Old Church Slavonic (Eckhoff and Berdicevskis, 2015)	Slavonic	1	800.0 – 1099.0	0.02×10^{7}	✗	
PROIEL-Old Church Slavonic (Haug and Jøhndal, 2008)	Slavonic	1	800.0 – 1099.0	0.01×10^{7}	✓	Eckhoff and Janda (2014)
TOROT-Kiev-era Old East Slavic (Eckhoff and Berdicevskis, 2015)	Slavonic	1	800.0 – 1250.0	0.09×10^{6}	✗	

Table 4: Summary of the 81 diachronic corpora surveyed by Jayatilleke et al. (2026); including and sorted by the language class as defined by Ranathunga and de Silva (2022).

	Primary Category		Secondary Category					Total
	Non-Fiction	Fiction	Religious	Poetry	Language	History	Medical	
5th	1	0	0	0	0	0	1	1
12th	0	1	0	1	0	0	0	1
13th	12	1	8	1	3	1	0	13
14th	2	2	3	0	0	1	0	4
15th	4	4	3	3	2	0	0	8
16th	1	0	0	1	0	0	0	1
18th	4	0	2	0	1	0	1	4
19th	7	3	4	3	2	1	0	10
20th	12	5	6	6	2	2	0	*16
Total	43	16	26	15	10	5	2	59

Table 5: Distribution of Books by Written Centuries and Genres in SDC.
*The total count for the secondary category in the 20th century is 16, while the overall number of books is 17. This discrepancy occurs because the book '*Hithopadhesha Sannaya*', which offers advice, was not classified under any of the five secondary categories. It is important to note that this distributional analysis is based on SiDiaC-v.2.0 (Jayatilleke et al., 2026), which serves as the foundation for SDC.

In these cases, it defaults to using the original word form when no explicit lemma is provided. To further ensure data integrity, we conducted a filtration step that scans each extracted lemma for validity, where any lemma that contains non-Sinhala characters is flagged as erroneous and replaced with the original word.

C.2 POS Tagging

During lemmatisation with SinMorphy, the tool assigned probable POS tags based on established morphological patterns, as detailed in Appendix **??**. However, there were instances where it assigned multiple possible POS tags. For example, the word 'කිරේ \ hɪre:' was tagged as either a noun or a verb. To determine the correct POS tags in these cases, we utilised the *TnT POS Tagger for Sinhala*[14] (Fernando and Ranathunga, 2018).

Trigrams'n'Tags (TnT) (Brants, 2000) is a statistical tool that employs the *Viterbi algorithm* (Forney, 2005) to implement second-order Markov models for POS tagging. It generates taggers based on *Hidden Markov Models* (HMMs) that calculate the most likely sequence of grammatical tags for a given text. To improve the accuracy of these probability estimates and effectively address data sparsity, the method uses linear interpolation as a smoothing technique.

The TnT (HMM) model was trained on two manually annotated datasets: a *News corpus* containing 200,000 words from various sections of local newspapers (such as general news, foreign news, and sports) and an *Official Document corpus* consisting of 83,000 words from government circulars, reports, and letters. Both corpora were annotated using the second level of the "*Comprehensive Multilevel Sinhala POS Tag Set* (Fernando et al., 2016)," which features 31 distinct tags designed to accommodate the vocabulary differences between the semi-spoken style of news and the formal written style of official documents.

During the implementation of the *TnT POS Tagger for Sinhala*, we had to make several adjustments to meet our requirements, where we improved the system to make it more robust and efficient. First, we faced the unavailability of the original C-based TnT tagger. To address this, we emulated the TnT behaviour using the `nltk.tag.tnt`[15] module. Secondly, we encountered an efficiency issue with the standard *Viterbi algorithm* (Forney, 2005) used in the model. This algorithm explores every possible path through the HMM states, resulting in a complexity of approximately $O(T^L)$ for a sentence of length L and tagset size T. To improve this, we subclassed '`nltk.tag.tnt.TnT`' and overrode the '`tag()`' method with an iterative *Beam Search algorithm*. In this mechanism, at each step for a word, we rank all possible partial paths based on their log-probability and retain only the top N best paths, determined by the beam width. We set the beam width to 100, which significantly reduces complexity to linear $O(L \cdot N \cdot T)$. This adjustment

[14] https://github.com/nlpcuom/Sinhala-POS-Tagger

[15] https://www.nltk.org/api/nltk.tag.tnt.html

allows for instantaneous tagging of long sentences without a significant loss in accuracy.

Due to the presence of textual content that spans across different time periods, the lexicon lacked certain words in the corpus, resulting in 23,180 words being labelled as '*UNK*'. To address this issue, we utilised another POS tagger from the `Sinling`[16] library to specifically identify the POS tags of the words that had been assigned multiple tags by `SinMorphy`.

To address ambiguities in POS tagging and morphological analysis, we employ a priority-based selection algorithm that integrates multiple data sources. For each token, we extract potential POS tags along with their corresponding morphological structures from `SinMorphy`. In cases where multiple POS candidates are given, the algorithm disambiguates the token by first consulting the *TnT* tagger, with the *Sinling* tagger serving as a fallback option.

The disambiguation process follows a strict hierarchy to ensure consistency between high-level POS categories and low-level morphological features. Initially, raw POS tags from secondary taggers such as `TnT` and `Sinling` are mapped to a standardised set of seven main-level categories, including *NOUN*, *VERB*, *PART*, *PROPER NOUN*, *ADJECTIVE*, *PRONOUN*, and *FOREIGN WORD*, while unidentified symbols are categorised as UNKNOWN. For every token, the morphological analyser provides a list of possible tags and a corresponding list of morphologies, which are maintained in parallel to ensure each tag remains linked to its correct linguistic interpretation.

The decision logic dictates that if only one candidate is available, it is automatically selected. In cases of ambiguity where multiple candidates exist, the algorithm searches for a match between the mapped `TnT` result and the candidate set. If the `TnT` result is unknown or does not match any candidate, the algorithm repeats this search using the mapped `Sinling` result. If no valid match is found from either tagger, the system defaults to the first candidate in the analyser's output to maintain data integrity. Once the final tag is determined, its specific index is used to retrieve the matching morphological string. The process concludes with lemma extraction, where the string is split at the first occurrence of the "+" delimiter to isolate the leading root substring.

[16] https://github.com/ysenarath/sinling

D Contextualised Embeddings

D.1 Continual Pre-training (CPT)

The applied CPT strategies were optimised for the distinct structural mechanisms of causal and masked language models. To evaluate the downstream impact of structural vocabulary adaptation across disparate model families, we selected `Llama-3.1-8B` and its downstream Sinhala variant `SinLlama` as our Causal Language Model (CLM) representative, and `XLM-R` as our Masked Language Model (MLM) representative. For both architectures, dynamic vocabulary expansion is initiated by extracting out-of-vocabulary (OOV) terms from the entire `SiDiaC-v.2.0` corpus via an optimised set-difference lookup. However, due to structural divergence in how autoregressive models construct representations versus how bidirectional encoders capture context, the subsequent optimisation, initialisation, and training strategies are decoupled into architecture-specific tracks.

For the CLM stream (`Llama-3.1` and `SinLlama`), the approach prioritises structural stabilisation to prevent representation collapse within the inner transformer blocks. Newly appended tokens were integrated into the network by initialising their weights with the exact average vector computed from the model's pre-existing embedding matrix. The adaptation then follows a decoupled two-stage paradigm. Stage 1 consists of a dedicated embedding warmup phase over a single epoch; all core attention layers are locked, and Low-Rank Adaptation (LoRA) matrices are applied exclusively to the input token embeddings and final language modelling heads to pull the new coordinates into a stable semantic space. Once boundary stability is achieved, these parameters are mathematically merged back into the base layers, and Stage 2 initiates deep contextual learning. In this final phase, the base model is memory-optimised via 4-bit quantisation, and a comprehensive LoRA configuration is deployed across all attention projection and Multi-Layer Perceptron (MLP) layers over 3 epochs with early stopping with a patience of 3 each 50 steps monitoring validation loss.

Conversely, the MLM stream (XLM-R) is adapted using an end-to-end, full-parameter optimisation loop to align the expanded token structures with its pre-existing multilingual representations. Unlike the autoregressive track, XLM-R's native architecture lacks parameter wrappers, enabling di-

rect structural surgery on its standard embedding layers. The model is expanded to its final terminal vocabulary size (totalling $307,386$ tokens after adding $57,384$ tokens) and trained directly on a MLM objective where a standard 15% token masking probability is enforced. Because this process tunes the model's bidirectional contextual weights globally, all encoder layers are unfrozen and optimised simultaneously.

D.2 Intrinsic Evaluation

To evaluate the linguistic processing efficiency of the models on `SDC` corpus, we calculate Perplexity (PPL) and Bits Per Byte (BPB) using a sliding window approach optimised for long sequences. For each sequence, we compute the total Negative Log-Likelihood (NLL) by shifting an evaluation window of size W by a scoring stride S. To isolate the evaluation to the target stride and prevent context bleeding from inflating the prediction quality, tokens belonging to the historical context window are explicitly masked by setting their target labels to -100. Because standard deep learning frameworks compute the cross-entropy loss as an average over active targets, we reconstruct the true segment NLL by scaling the model's average loss output by the precise count of non-masked target tokens within that window, denoted as $N_{\text{active}} = \sum \mathbb{I}(\text{target}_j \neq -100)$. Accumulating these values over the entire evaluation corpus, the final corpus-level metrics are computed as:

$$\text{PPL} = \exp\left(\frac{\sum_k \text{NLL}_k}{\sum_k N_{\text{active},k}}\right) \tag{2}$$

$$\text{BPB} = \frac{\sum_k \text{NLL}_k}{B_{\text{total}} \cdot \ln(2)} \tag{3}$$

where NLL_k and $N_{\text{active},k}$ represent the absolute negative log-likelihood and active target count for the k-th sliding window segment, respectively, and B_{total} is the absolute byte length of the raw, UTF-8 encoded text corpus. While perplexity remains inherently dependent on the specific tokenisation vocabulary size due to its per-token normalisation, the derivation of `BPB` yields a highly reliable, tokeniser-agnostic benchmark that enables direct cross-architectural evaluation.

The impact of tokenisation on perplexity scores is clearly illustrated in Table 1 when comparing the results of fine-tuned models versus base models. In the conducted CPT experiments, where we first expanded the vocabulary, we observed significant disparity of perplexity scores compared to the standard versions of the corresponding models. A larger vocabulary provides a greater range of options for selecting the next token, which spreads out the model's probability distributions. As a result, the model faces more difficulty in confidently choosing the correct option, leading to increased uncertainty and higher perplexity scores. Unfortunately, the BPB scores also increased, though not as significantly as the perplexity scores after conducting CPT. This indicates a need for more data for this type of task, even after utilising the entirety of `SiDiaC-v.2.0`, which comprises approximately 241k words.

In terms of decoder-only models, `Llama-3.1-8B` outperforms others when considering both metrics. However, the performance of `SinLlama`, which is a continually pre-trained version of `Llama-3.1-8B` trained on a 10 million corpus with an expanded tokeniser (Aravinda et al., 2025), highlights the impact of tokeniser dependency on perplexity. This improvement is evident based on the superior BPB scores compared to the standard version. Following the approach by Wang et al. (2024) for using instruct models to generate embeddings, we experimented with the instruct variant of `Llama-3.1-8B`. Unfortunately, this led to lower scores in both perplexity and BPB compared to the base model.

In our analysis, the only encoder-only model we considered, `XLM-R`, achieved the best BPB score. However, because the MLM objective computes loss for each isolated masked token, it is unable to generate perplexity scores. Given that the BPB score, which is tokeniser-agnostic, proved to be more important and produced superior results, we concluded that `XLM-R` is the model to use for the bidirectional semantic impact scoring evaluation discussed in section 4.2.

E Computational Semantic Mapping

Two semantic mapping approaches for `Word2Vec` and `FastText` were utilised for century-wise semantic analysis, as discussed in section 3.3. The alignment mechanisms are summarised and illustrated in Figure 4.

E.1 Orthogonal Procrustes Alignment

The OP alignment was conducted using two distinct approaches: analytical and neural, as described in subsection 3.3.2. We evaluated the alignment

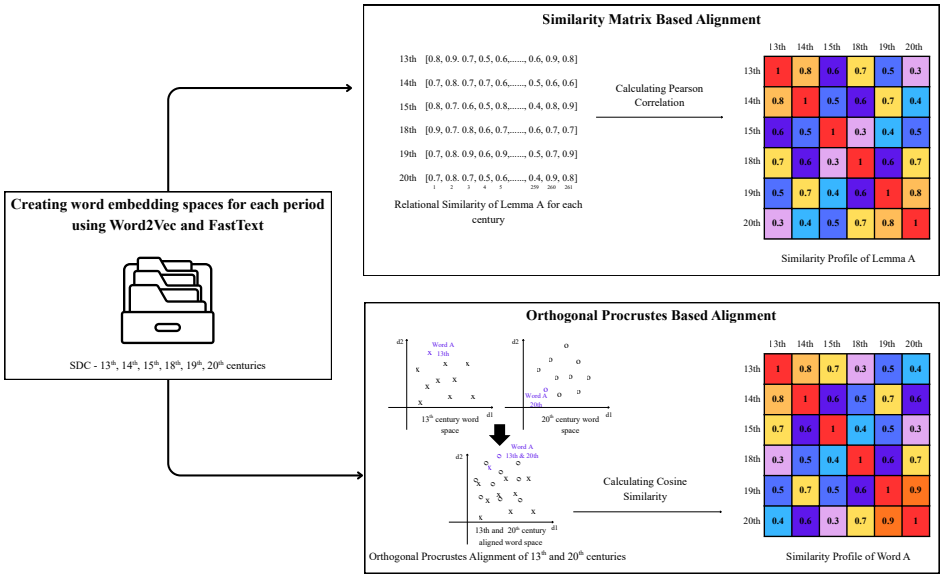


Figure 4: A framework for analysing diachronic semantic shifts by creating periodic embedding spaces using `Word2Vec` and `FastText`, followed by Similarity Matrix-based and Orthogonal Procrustes-based alignment.
† Note that Orthogonal Procrustes-based alignment was conducted through two approaches: (1) aligning two word spaces to one, as depicted in this illustration, (Neural Procrustes Alignment); and (2) aligning all embedding spaces with the 13th century as the global reference, (Analytical Procrustes Alignment).

approaches based on two metrics: Procrustes disparity and rotational distance. In this section, we will examine the metric results obtained for each model-alignment combination corresponding to the alignment from the reference century to the target century. This analysis clarifies the average scores presented in Table 2.

Target Century	Reference Century	Procrustes Disparity	Rotational Distance
14th	13th	172.37	9.93
15th	13th	173.04	10.09
18th	13th	168.95	9.83
19th	13th	177.85	9.82
20th	13th	178.58	9.98

Table 6: Evaluation of Analytical Procrustes Alignment Performance of `Word2Vec` embedding spaces.

Target Century	Reference Century	Procrustes Disparity	Rotational Distance
14th	13th	153.12	9.14
15th	13th	159.41	9.30
18th	13th	140.86	8.77
19th	13th	165.56	9.22
20th	13th	166.86	9.35

Table 7: Evaluation of Neural Procrustes Alignment Performance of `Word2Vec` embedding spaces.

Target Century	Reference Century	Procrustes Disparity	Rotational Distance
14th	13th	59.73	22.75
15th	13th	56.81	22.82
18th	13th	69.72	22.63
19th	13th	51.91	22.97
20th	13th	49.68	22.95

Table 8: Evaluation of Analytical Procrustes Alignment Performance of `FastText` embedding spaces.

Target Century	Reference Century	Procrustes Disparity	Rotational Distance
14th	13th	59.15	19.71
15th	13th	56.33	19.79
18th	13th	68.38	19.47
19th	13th	51.80	19.72
20th	13th	49.62	19.72

Table 9: Evaluation of Neural Procrustes Alignment Performance of `FastText` embedding spaces.

Our analysis reveals a fundamental theoretical conflict in how data density affects vector space stability across the two static embedding architectures, as detailed across Figure 1 and Tables 6–8. Standard machine learning theory dictates that increased training data should improve structural stability and alignment precision. This expected behaviour is observed clearly in the fine-tuned `FastText` models. As data density increases from the 18th century (4,707 tokens) to the 20th century (17,295 tokens), the Procrustes Disparity drops significantly from its peak error of 68.38 down to its global minimum of 49.62 in Table 8.

In contrast, `Word2Vec` exhibits the exact opposite trend, providing results that appear theoretically incorrect. When data volume increases from the 18th to the 20th century, `Word2Vec`'s metrics degrade simultaneously: its Procrustes Disparity rises from 140.86 to its absolute maximum of 166.86, while its Rotational Distance increases from 8.77 to a peak of 9.35, as shown in Table 7 A similar trend is noted in Table 6 regarding Procrustes Disparity,

though the Rotational Distance appears somewhat scattered, showing no clear trend. This means that instead of converging toward the 13th-century baseline with more text data, the `Word2Vec` space shifts further away and requires more global coordinate rotation to achieve an alignment compromise. In a from-scratch word-level model, scaling up the token count introduces a severe vocabulary expansion problem. Rather than simply refining existing word paths, extra data introduces thousands of completely unique, rare word context windows. This forces the high-dimensional space to expand and distort dynamically to fit these new parameters. Consequently, when forced back onto the 13th-century anchor via a rigid linear transformation matrix Q, the true semantic drift and vocabulary expansion accumulated over 7 centuries are fully exposed, yielding a higher disparity and requiring greater rotational force.

Conversely, `FastText` avoids this distortion because it is fine-tuned. Its underlying subword manifolds are bound by prior geometric constraints that dictate where words must reside. Additional tokens simply resolve and smooth out the pre-existing coordinate space rather than chaotically expanding it. This enables a gradual reduction in disparity as more data is added, while maintaining a stable rotational distance at approximately 19.72 across all centuries, as shown in Table 8. Similar trends can be observed in Table 9. Thus, while `Word2Vec` fails to provide theoretically good alignment metrics, its rising scores are an accurate, unconstrained mathematical footprint of the actual structural evolution of a language learned from scratch.

The higher Rotational Distance in `FastText` ($\approx$ 19.72 in Table 8) compared to `Word2Vec` ($\approx$ 9.35 in Table 7) is caused by how the models are initialised. Because `Word2Vec` is learned from scratch, its coordinate system is highly flexible. It easily rotates locally to match small vocabularies with very little structural effort.

In contrast, `FastText` inherits a rigid, pre-trained subword structure. This pre-existing manifold binds the entire language space together as a single block, preventing it from collapsing or shifting easily. Aligning this large, pre-structured space to the historical anchor requires a much larger global rotation. While this requires more rotational effort, it ensures that the internal structure of the language remains intact, ultimately yielding vastly superior alignment precision.

F Evaluation of Semantic Drift

F.1 Statistical Neighbourhood Analysis

There are six possible combinations of models used to generate embeddings and the mechanisms for semantic mapping. Each of these model-to-mapping combinations features 15 different alignments in word space. The frequency distribution of lemmas (used in the SMA approach) or words (used in the OP approaches), which show similarity and decline according to the details outlined in subsection 4.1, is illustrated in Table 10.

Subsequently, we conducted a neighbourhood analysis of the lemmas or words that have undergone semantic drift, following the approach detailed in subsection 4.1. It is important to note that this is not a Bag-of-Words (BoW) analysis, such as the one conducted by Jayatilleke et al. (2026) on the same subset of `SiDiaC-v.2.0` considered in this study. This neighbourhood analysis focuses on the top 5 similar words that remain closest within the latent space. By examining the related meanings of neighboring words, as described in Soratha Thera (2011) and using an interactive Sinhala dictionary, Wahara[17], we aimed to disambiguate the possible meanings of the lemmas\words that have experienced semantic drift.

Based on the six model-to-mapping combinations mentioned, we examine three intriguing examples of semantic drift for each. For the `Word2Vec` embeddings using the SMA approach, the selected examples are ‘පුරුෂ \ pʊrʊʃʌ’, ‘චන්ද්‍රයා \ tʃʌnðrʌjɑː’, and ‘සැප \ sæpʌ’, as shown in Table 11.

The lemma ‘පුරුෂ \ pʊrʊʃʌ’ in the 18th century was associated with descriptors highlighting the extraordinary physical attributes of the Supreme Buddha. Examples include ‘කේතුමාලා \ keːθʊmɑːlɑː’, which means the shining rays emanating from the Buddha’s head, and ‘අසූ \ ʌsuː’, referring to the number eighty, which relates to ‘ [18]’, the eighty distinct physical characteristics of the Buddha. Additionally, the term ‘ප්‍රභානු \ keːθʊmɑːlɑː’ signifies light, radiance, and brilliance, again connected to the physical attributes of the Buddha. By the 20th century, however, the word underwent a significant process of secularisation. It lost its ties to the supernatural attributes of the Buddha and instead began to reflect the ordinary human experience of mortality and suffering. Its semantic neighbours depict the raw realities of the ordinary individual

[17] http://crawler.nlpc.uom.lk/

[18] https://pitaka.lk/books/pujawaliya/11-6.html

	Mapping	13↔14	13↔15	13↔18	13↔19	13↔20	14↔15	14↔18	14↔19	14↔20	15↔18	15↔19	15↔20	18↔19	18↔20	19↔20	Total
Word2Vec	SMA	13	12	17	16	17	13	16	16	14	20	12	14	16	19	14	229
	Analytical OP	39	39	45	43	43	37	35	38	43	37	48	46	49	46	47	635
	Neural OP	37	47	43	47	38	39	43	43	49	39	47	46	47	39	44	648
FastText	SMA	6	1	8	4	2	6	6	3	3	6	0	1	5	7	1	59
	Analytical OP	17	19	18	12	13	19	17	17	22	20	18	15	23	19	22	271
	Neural OP	16	19	18	12	14	18	19	19	22	21	19	15	20	21	21	274

Table 10: Frequency distribution of lemmas\words exhibiting significant similarity dips across century-wise aligned semantic spaces (Word2Vec vs. FastText).

Lemma	Similarity	Ref Century	Ref Century Neighbors	Target Century	Target Century Neighbors
පුරුෂ \ pʊrʊʃʌ	0.0274	18th	හාවිතානු \ haːviθaːnʊ, අසූ \ ʌsuː, කේතුමාලා \ keːθʊmaːlaː, ප්‍රභානු \ keːθʊmaːlaː, පූර්වාංගුලි \ pʊrvaːngʊlɪ	20th	ජරා \ dʒʌraː, මරණ \ mʌrʌnʌ, මිදුනු \ mɪðʊnʊ, දැන්මද \ ðænmʌðʌ, එකක්හු \ ɛkkʌhʊ
චන්ද්‍රයා \ tʃʌnðrʌjaː	0.1379	14th	උදාකරවා \ ʊðaːkʌrʌvaː, යන්හි \ jʌnhɪ, ස්ථාන \ sθʰaːnʌ, වෛනෙජන \ vaɪnɛdʒʌnʌ, කෞමද \ kaʊmʌðʌ	15th	ධීනෘවූ \ ðʰiːθruːvuː, පුණ්ණ \ pʊnnʌ, උලකුඩය \ ʊlʌkʊdʌjʌ, දූ \ ðuː, දේවී \ ðeːviː
සැප \ sæpʌ	0.1755	14th	දුක් \ ðʊk, වටනේය \ vʌtʌneːjʌ, සසරෙහි \ sʌsʌrɛhɪ, අමෘත \ ʌmʊrθʌ, ව්‍යාධි \ vyaːðiː	15th	සෞඛ්‍ය \ saʊkyʌ, ප්‍රවෘත්ත \ prʌvʊrθʌ, මිත්‍රොන්තමයෝ \ mɪθrɒnθʌmʌjɔː, දෙකෙහිම \ ðɛkɛhɪmʌ, තුමෝ \ θʊmɔː

Table 11: Three lexical items exhibiting semantic drift, analysed using Word2Vec embeddings and Similarity Matrix Alignment (SMA).

(එකක්හු \ ɛkkʌhʊ), bound to a deteriorating physical existence (ජරා \ dʒʌraː) and death (මරණ \ mʌrʌnʌ).

The lemma 'චන්ද්‍රයා \ tʃʌnðrʌjaː' usually means the moon. However, in the 14th century, it appears in a religious context where it is linked with words like 'උදාකරවා \ ʊðaːkʌrʌvaː', which metaphorically refers to the spreading of the Buddha's teachings as spiritual light across places. It also co-occurs with 'කෞමද \ kaʊmʌðʌ', meaning a white lily or lotus, which is associated with purity and spirituality. By the 15th century, the meaning of 'චන්ද්‍රයා \ tʃʌnðrʌjaː' changes. It moves away from religious use and becomes more connected to courtly and romantic language. In this period, it is often associated with royal and noble female terms such as 'ධීනෘවූ \ ðʰiːθruːvuː', 'දූ \ ðuː', and 'දේවී \ ðeːviː', where the moon is used as a symbol of beautiful women and royal identity.

The core meaning of the lemma 'සැප \ sæpʌ' is happiness or comfort. In the 14th century, its closest semantic neighbours show a very spiritual and religious understanding. In this context, true happiness is not ordinary pleasure, but the complete ending of suffering in samsara, linked with words like දුක් \ ðʊk and සසරෙහි \ sʌsʌrɛhɪ. It is also connected to 'අමෘත \ ʌmʊrθʌ', meaning deathless Nibbāna, and 'ව්‍යාධි \ vyaːðiː', meaning sickness or illness, which represents the suffering from which liberation is achieved. By the 15th century, the meaning of the lemma shifts away from this spiritual sense and becomes more connected to everyday life. Happiness is no longer understood in a spiritual manner, but is instead linked with physical well-being (සෞඛ්‍ය \ saʊkyʌ) and warm social relationships (මිත්‍රොන්තමයෝ \ mɪθrɒnθʌmʌjɔː).

For the Word2Vec embeddings using the analytical procrustes alignment approach, the three selected examples are 'රූප \ ruːpʌ', 'සිත \ sɪθʌ', and 'මල් \ mʌl' as shown in Table 12.

In the 13th century, the word 'රූප \ ruːpʌ' is associated with terms describing physical appearance and bodily form. Its semantic neighbours cluster around physical grace (විලාස \ vɪlaːsʌ, විලාසයෙනුදු \ vɪlaːsʌjɛnʊðʊ) and radiant visual beauty or splendour (ශොභායි \ ʒɒbʰaːj), referring to brightness, brilliance, and aesthetic attractiveness. By the 19th century, the lemma shifts into a more technical and philosophical usage within Buddhist thought, where it denotes "form" or "matter." In this later stage, its semantic neighbours belong to abstract doctrinal categories such as the five aggregates (ස්කන්ධයන් \ skʌnðʰʌjʌθ, meaning the aggregates that constitute a being), සංඛාර \ sʌŋkʰaːrʌ (understood as conditioned phenomena encompassing all formations arising through karmic and causal processes, both wholesome and unwholesome), mental factors (චෛතසික \ tʃaɪθʌsɪkʌ), and the teachings of the Buddha (ධර්මයොන් \ ðʰʌrmʌjɒθ).

In the 13th century, the word 'සිත \ sɪθʌ' is associated with a semantic field related to intention and basic mental activity. Its closest neighbours include terms such as prior intention or disposition (පෙරදැරිව \ pɛrʌðærɪvʌ), action or engagement (දැමිය \ ðæmɪjʌ), and ground or context (බිම් \ bɪm), along with unwholesome action (අකුසලට \ ʌkʊsʌlʌtʌ), suggesting an early connection between the mind and moral behaviour. By the 18th century, the lemma shifts into a more physical

Word	Similarity	Ref Century	Ref Century Neighbors	Target Century	Target Century Neighbors
රූප \ ruːpʌ	0.4665	13th	විලාස \ vɪlaːsʌ, ප්‍රජ්වලික \ prʌdʒvʌlɪθʌ, ශොභායි \ ʒɒbʰaːj, විලාසයෙනුදු \ vɪlaːsʌjɛnʊðʊ, අශීනි \ ʌʒiːθɪ	19th	ධර්මයොත් \ ðʰʌrmʌjɒθ, අයිතිය \ ʌjɪθɪjʌ, ස්කන්ධයන් \ skʌnðʰʌjʌθ, සංඛාර \ sʌŋkʰaːrʌ, චෛතසික \ tʃaɪθʌsɪkʌ
සින \ sɪθʌ	0.4876	13th	පෙරදැරිව \ pɛrʌðærɪvʌ, දැමිය \ ðæmɪjʌ, හම්ය \ bʰʌmjʌ, බිම් \ bɪm, අකුසලට \ ʌkʊsʌlʌtʌ	18th	සන්නිපාතජ්වර \ sʌnnɪpaːθʌdʒvʌrʌ, සුදුවීම \ suðʊviːmʌ, ඇස \ æsʌ, ශ්ලෙස්මාවර \ ʒlɛsmaːvʌrʌ, ජ්වරනාප \ dʒvʌrʌθaːpʌ
මල් \ mʌl	0.5788	14th	වදාරන්නේ \ vʌðaːrʌnne, පසඟ \ pʌsʌngʌ, ගැත්තවුන් \ gæθθʌvʊn, දවසට \ ðʌvʌsʌtʌ, බන \ bʌnʌ	18th	කිණිහිරි \ kɪnɪhɪrɪ, දහවත්තරු \ ðʌhʌvʌθθʊrʊ, බෙරලිය \ bɛrʌlɪjʌ, සුදු \ suðʊ, වුවමනාතන්හි \ vʊvʌmʌnaːθʌnhɪ

Table 12: Three lexical items exhibiting semantic drift, analysed using Word2Vec embeddings and Analytical Procrustes Alignment.

and pathological semantic field. Its neighbours include terms related to illness and bodily conditions, such as a type of fever involving accumulation of symptoms (සන්නිපාතජ්වර \ sʌnnɪpaːθʌdʒvʌrʌ), paleness (සුදුවීම \ suðʊviːmʌ), eye (ඇස \ æsʌ), phlegm-related condition (ශ්ලෙස්මාවර \ ʒlɛsmaːvʌrʌ), and fever/heat (ජ්වරනාප \ dʒvʌrʌθaːpʌ). In this period, 'සින \ sɪθʌ' is linked not only to mental processes but also to bodily states and health-related interpretations of experience.

In the 14th century, the word 'මල් \ mʌl' appears in a context related to religious listening and devotion. Its neighbours include speaking or preaching (වදාරන්නේ \ vʌðaːrʌnne), five ritual elements (පසඟ \ pʌsʌngʌ), devotees (ගැත්තවුන් \ gæθθʌvʊn), and Buddhist preachings (බන \ bʌnʌ). In this context, 'මල් \ mʌl' refers not only to flowers as physical objects, but also to their use as offerings in worship and in the setting of listening to Buddhist preachings. By the 18th century, the lemma shifts into a more natural and descriptive context. Its neighbours include plant names such as කිණිහිරි \ kɪnɪhɪrɪ and බෙරලිය \ bɛrʌlɪjʌ, and the colour term 'සුදු \ suðʊ' (white). In this period, 'මල් \ mʌl' is mainly connected to flowers as part of the natural environment, rather than religious or ritual use.

For the Word2Vec embeddings using the neural procrustes alignment approach, the three selected examples are 'රන් \ rʌn', 'මහත් \ mʌhʌθ', and 'බලයෙන් \ bʌlʌjɛn' as shown in Table 13.

In the 19th century, the word 'රන් \ rʌn' appears in a semantic environment connected with material objects and social actions. Its neighbours include containers or vessels (භාජනයෝ \ bʰaːdʒʌnʌjɔː), food such as rice (බත් \ bʌθ), personal reference (හෙතෙම \ hɛθɛmʌ), giving or offering (ප්‍රදානය \ prʌðaːnʌjʌ), and leaves or plates (පත්‍රයෝ \ pʌθrʌjɔː). In this context, 'රන් \ rʌn' is associated with tangible items used in everyday or ritual transactions, especially objects involved in giving and offering. By the 20th century, the word shifts into a more poetic and ornamental semantic field. Its neighbours include decorative or natural elements such as a creeper or vine (ලතාවක් \ lʌθaːvʌk), blooming or fullness (සුපුල් \ supʊl), the hand or palm (සුරතෙ \ suːrʌθɛ), binding or tying (බන්දනය \ bʌnðʌnʌjʌ), and bringing or leading (ගෙන්නමින් \ gɛnnʌmɪn). In this period, 'රන් \ rʌn' is used more in an aesthetic and figurative sense, often linked with beauty, decoration, and poetic imagery rather than material exchange.

In the 15th century, the word 'මහත් \ mʌhʌθ' appears in a semantic environment associated with intellectual and learned qualities. Its neighbours include terms such as attained or gained (ලද්දා \ lʌððʌ), learned or educated (වියත් \ vɪjʌθ), realised or obtained (අධිගත \ ʌðʰɪgʌθʌ), scholar (පණ්ඩිත \ pʌndɪθʌ), and published or expressed (පලකළ \ pʌlʌkʌlʌ). In this context, 'මහත් \ mʌhʌθ' is linked with ideas of greatness in knowledge, learning, and intellectual achievement. By the 19th century, the word shifts into a more everyday and informal semantic field. Its neighbours include disturbance or confusion (කලබල \ kʌlʌbʌlʌ), aside or away (අහක \ ʌhʌkʌ), self-praise or arrogance (අත්තුක්කංසන \ ʌθθʊkkʌŋsʌnʌ), and many or large quantities (ගොඩක් \ gɒdʌk). In this period, 'මහත් \ mʌhʌθ' is used more in relation to quantity, intensity, or everyday descriptive meanings rather than scholarly or intellectual greatness.

In the 15th century, the word 'බලයෙන් \ bʌlʌjɛn' appears in a semantic environment associated with force and physical action. Its neighbours include whirlpool or swirling water (දියසුලි \ ðɪjʌsʊlɪ), swirling or rotational motion (සළාදී \ sʌlaːðiː), attack or strike (ප්‍රහාර \ prʌhaːrʌ), and rising or emerging (සෑදෙමින්නැඟි \ sæːðɛmɪnnæŋgɪ). In this context, 'බලයෙන් \ bʌlʌjɛn' is linked with the idea of physical force, impact, and strong natural or dynamic motion. By the 20th century, the word

Word	Similarity	Ref Century	Ref Century Neighbors	Target Century	Target Century Neighbors
රන් \ rʌn	0.2549	19th	භාජනයෝ \ bʰɑːdʒʌnʌjɔː, බත් \ bʌθ, හෙතෙම \ hɛθɛmʌ,	20th	ලනාවක් \ lʌθɑːvʌk, සුපුල් \ sʊpʊl, සුරතෙ \ sʊrʌθɛ,
මහත් \ mʌhʌθ	0.2775	15th	ලද්දා \ lʌðdʌ, වියත් \ vijʌθ, අධිගත \ ʌðʰigʌθʌ, පණ්ඩිත \ pʌndɪθʌ, පලකළ \ pʌlʌkʌlʌ	19th	කලබල \ kʌlʌbʌlʌ, අටේ \ ʌteː, අහක \ ʌhʌkʌ, අත්තුක්කංසන \ ʌθθʊkkʌŋsʌnʌ, ගොඩක් \ gɒdʌk
බලයෙන් \ bʌlʌjɛn	0.371	15th	දියසුලි \ ðɪjʌsʊlɪ, ඡෂ්ටා \ tʃʰʌʃtɑː, සළාදී \ sʌlɑːðiː, ප්‍රහාර \ prʌhɑːrʌ, සෘදෙමින්නැඟි \ sæːðɛmɪnnæŋgɪ	20th	මන්ත්‍ර \ mʌnθrʌ, යකුන් \ jʌkʊn, පරිවය \ pʌrɪtʃʌjʌ, සහචර \ sʌhʌtʃʌrʌ, හටගත් \ hʌtʌgʌθ

Table 13: Three lexical items exhibiting semantic drift, analysed using `Word2Vec` embeddings and Neural Procrustes Alignment.

shifted into a more supernatural and social semantic field. Its neighbours include spells or chants (මන්ත්‍ර \ mʌnθrʌ), demons or spirits (යකුන් \ jʌkʊn), practice or experience (පරිචය \ pʌrɪtʃʌjʌ), companions or associates (සහචර \ sʌhʌtʃʌrʌ), and arising or coming into being (හටගත් \ hʌtʌgʌθ). In this period, 'බලයෙන් \ bʌlʌjɛn' is associated more with supernatural power, ritual force, and socially embedded notions of influence rather than purely physical strength.

For the `FastText` embeddings using the SMA approach, the three selected examples are 'අග්‍ර \ ʌgrʌ', 'උපද \ ʊpʌðʌ', and 'භාව \ bʰɑːvʌ' as shown in Table 14.

In the 13th century, the lemma 'අග්‍ර \ ʌgrʌ' appears in a semantic environment associated with spiritual and cosmological ideas. Its neighbors include the wish-fulfilling tree (කප්රුක \ kʌprʊkʌ), enlightenment or Buddhahood (බුදුවී \ bʊðʊviː), name of a monk (චක්ඛුපාල \ tʃʌkkʰʊpɑːlʌ), universality or "all" (සියල්ල \ sɪjʌllʌ), and mental activity (සිතින් \ sɪθɪn). In this context, 'අග්‍ර \ ʌgrʌ' is linked with ideas of highest excellence, spiritual attainment, and supreme mental or religious states. By the 18th century, the lemma shifts into a more physical and bodily semantic field. Its neighbours include mouth or face (මුඛ \ mʊkʰʌ), two parts or divisions (දෙඅංශ \ ðɛʌŋʒʌ), divine or celestial (දිව්‍ය \ ðɪvyʌ), teeth (දන්ත \ ðʌnθʌ), and work or service for others (පරවැඩෙහි \ pʌrʌvæɖɛhɪ). In this period, 'අග්‍ර \ ʌgrʌ' is more associated with physical parts, bodily description, and social or divine attributes rather than purely spiritual or abstract excellence.

In the 13th century, the word 'උපද \ ʊpʌðʌ' appears in a semantic environment associated with life, destiny, and human existence. Its neighbors include astrologer or fortune teller (දෛවඥ \ ðaɪvʌnyʌ), death (මරණ \ mʌrʌnʌ), king (රාජයා \ rɑːdʒʌjɑː), worship or offering (පූද \ pʊðʌ), and life (ජීවිතය \ dʒiːvɪθʌjʌ). In this context, 'උපද \ ʊpʌðʌ' is connected with ideas of birth, fate, and the cycle of human existence. By the 18th century, the lemma shifts into a more physical and material semantic field. Its neighbors include left side (වම් \ vʌm), bodily elements or substances (ධාතු \ ðʰɑːθʊ), well-being or comfort (සුවයෙන් \ sʊvʌjɛn), heavenly city or divine realm (දෙවිනුවර \ ðɛvnʊvʌrʌ), and negation or uncertainty (නොවෙද්ද \ nɒvɛððʌ). In this period, "උපද \ ʊpʌðʌ" becomes associated more with bodily, spatial, and worldly concepts rather than themes of destiny and existence.

In the 13th century, the lemma 'භාව \ bʰɑːvʌ' appears in a semantic environment associated with spiritual and philosophical ideas. Its neighbours include groups or sets of five (පඤ්චක \ pʌntʃʌkʌ), cosmic age or aeon (කල්පය \ kʌlpʌjʌ), proper establishment or calming of the mind (සමාධාන \ sʌmɑːðʰɑːnʌ), that which is beyond sensory perception or difficult to comprehend (අවිෂය \ ʌvɪʃʌjʌ), and Bodhisattvas (බෝසත්හු \ bɔːsʌθhʊ). In this context, 'භාව \ bʰɑːvʌ' is connected with religious states, spiritual development, and doctrinal thought. By the 18th century, the lemma shifts into a more ethical and practical semantic field. Its neighbours include radiant or brightly shining (විරාජිත \ vɪrɑːdʒɪθʌ), practice or conduct (ප්‍රතිපත්ති \ prʌθɪpʌθθɪ), uncertainty or negation (නොවෙද්ද \ nɒvɛððʌ), bodily joints or points of connection (සන්ධි \ sʌnðʰɪ), and good character or proper behaviour (සුචරිත \ sʊtʃʌrɪθʌ,). In this period, 'භාව \ bʰɑːvʌ' becomes more associated with moral behaviour, conduct, and practical human qualities rather than abstract spiritual or cosmological ideas.

For the `FastText` embeddings using the analytical procrustes alignment approach, the three selected examples are 'මාර්ග \ mɑːrgʌ', 'පෙර \ pɛrʌ', and 'සමියක් \ sʌmjʌk' as shown in Table 15.

In the 13th century, the word 'මාර්ග \ mɑːrgʌ' appears in a semantic environment associ-

Lemma	Similarity	Ref Century	Ref Century Neighbors	Target Century	Target Century Neighbors
අග \ ʌgrʌ	0.2175	13th	කප්රුක \ kʌprʊkʌ, බුදුවී \ bʊðʊviː , චක්බුපාල \ tʃʌkkʰʊpɑːlʌ , සියල්ල \ sɪjʌllʌ ,සිතින් \ sɪθɪn	18th	මුඛ \ mʊkʰʌ , දෙඅංශ \ ðɛʌŋʒʌ , දිව්‍ය \ ðɪvyʌ , දන්න \ ðʌnθʌ , පරවැඩෙහි \ pʌrʌvædɛhɪ
උපද \ ʊpʌðʌ	0.3253	13th	දෛවඥ \ ðʌɪvʌnyʌ, මරණ \ mʌrʌnʌ, රාජ්‍යය \ rɑːdʒʌjɑː, පුද \ pʊðʌ, ජීවිතය \ dʒiːvɪθʌjʌ	18th	වම් \ vʌm , ධාතු \ ðʰɑːθʊ , සුවයෙන් \ sʊvʌjɛn , දෙවිනුවර \ ðɛvnʊvʌrʌ , නොවෙද්ද \ nɒvɛðdʌ
භාව \ bʰɑːvʌ	0.4127	13th	පඤ්චක \ pʌntʃʌkʌ, කල්පය \ kʌlpʌjʌ, සමාධාන \ sʌmɑːðʰɑːnʌ, අවිෂය \ ʌvɪʃʌjʌ, බෝසත්හු \ bɔːsʌθhʊ	18th	විරාජිත \ vɪrɑːdʒɪθʌ, ප්‍රතිපත්ති \ prʌθɪpʌθθɪ, නොවෙද්ද \ nɒvɛðdʌ, සන්ධි \ sʌnðʰɪ, සුචරිත \ sʊtʃʌrɪθʌ

Table 14: Three lexical items exhibiting semantic drift, analysed using `FastText` embeddings and Similarity Matrix Alignment (SMA).

Word	Similarity	Ref Century	Ref Century Neighbors	Target Century	Target Century Neighbors
මාර්ග \ mɑːrgʌ	0.6561	13th	මොහයෙන් \ mɒhʌjɛn, පිළිවෙළද \ pɪlɪvɛlʌðʌ, ආදරයෙන් \ ɑːðʌrʌjɛn, වැඩියෙන් \ vædɪjɛn, පිළිසිඳ \ pɪlɪsɪnðʌ	18th	කඳක් \ kʌnðʌk, පුබ්බ \ pʊbbʌ, ලැබුවා \ læbʊvɑː, පෙර \ pɛrʌ, දරුවන්ගේ \ ðʌrʊvʌngeː
පෙර \ pɛrʌ	0.7222	14th	කාලය \ kɑːlʌjʌ, ග්‍රහණය \ grʌhʌnʌjʌ, දවස \ ðʌvʌsʌ, මුළුල්ලෙහි \ mʊlʊllɛhɪ, ප්‍රවෘත්තිය \ prʌvʊrθɪjʌ	15th	පලු \ pʌlʊ, යැද \ jæðʌ, රජහුගේ \ rʌdʒʌhʊgeː, ගැබ \ gæbʌ, කළාක් \ kʌlɑːk
සම්යක් \ sʌmjʌk	0.7779	14th	වටනේ \ vʌtʌneː, අසවු \ ʌsʌv, රූපය \ ruːpʌjʌ, මහණෙක් \ mʌhʌnɛk, ඥාණය \ nyɑːnʌjʌ	20th	පෙළපතකින් \ pɛlʌpʌθʌkɪn, ක්‍රිස්තියානි \ krɪsθɪjɑːnɪ, ක්‍රියාවෙන් \ krɪjɑːvɛn, භාෂාවෙන් \ bʰɑːʃɑːvɛn, නූගතුන් \ nuːgʌθʊn

Table 15: Three lexical items exhibiting semantic drift, analysed using `FastText` embeddings and Orthogonal Procrustes Alignment.

ated with guidance, order, and progression. Its neighbours include in delusion or ignorance (මොහයෙන් \ mɒhʌjɛn), order or arrangement (පිළිවෙළද \ pɪlɪvɛlʌðʌ), with affection or love (ආදරයෙන් \ ɑːðʌrʌjɛn), in greater degree (වැඩියෙන් \ vædɪjɛn), and conception or arising (පිළිසිඳ \ pɪlɪsɪnðʌ). In this context, 'මාර්ග \ mɑːrgʌ' is linked with ideas of a path as a structured way of conduct, moral direction, or spiritual progression. By the 18th century, the word shifts into a more physical and textual semantic field. Its neighbours include the main body of a person excluding the head and limbs (කඳක් \ kʌnðʌk), a section or division of a book or collection of texts (පුබ්බ \ pʊbbʌ), received or obtained (ලැබුවා \ læbʊvɑː), before (පෙර \ pɛrʌ), and children (දරුවන්ගේ \ ðʌrʊvʌngeː). In this period, 'මාර්ග \ mɑːrgʌ' is more associated with physical movement, textual organisation, and everyday relational contexts rather than abstract moral or spiritual pathways.

In the 14th century, the word 'පෙර \ pɛrʌ' appears in a semantic environment associated with time and temporal reference. Its neighbors include time (කාලය \ kɑːlʌjʌ), grasping or understanding (ග්‍රහණය \ grʌhʌnʌjʌ), day (දවස \ ðʌvʌsʌ), throughout or entirely (මුළුල්ලෙහි \ mʊlʊllɛhɪ), and news or event (ප්‍රවෘත්තිය \ prʌvʊrθɪjʌ). In this context, 'පෙර \ pɛrʌ' is linked with ideas of time, sequence, and temporal framing of events. By the 15th century, the word shifts into a more narrative and situational semantic field. Its neighbours include fruit or leaf (පලු \ pʌlʊ), invited or requested through prayer (යැද \ jæðʌ), of the king (රජහුගේ \ rʌdʒʌhʊgeː), womb or inner space (ගැබ \ gæbʌ), and done or made (කළාක් \ kʌlɑːk). In this period, 'පෙර \ pɛrʌ' is used more within storytelling contexts and event descriptions rather than purely as a temporal marker.

In the 14th century, the word 'සම්යක් \ sʌmjʌk' appears in a semantic environment associated with correctness, knowledge, and spiritual discipline. Its neighbours include worthy or valuable (වටනේ \ vʌtʌneː), hearing or listening (අසවු \ ʌsʌv), form or appearance (රූපය \ ruːpʌjʌ), knowledge or insight (ඥාණය \ nyɑːnʌjʌ), and a monk (මහණෙක් \ mʌhʌnɛk). In this context, 'සම්යක් \ sʌmjʌk' is linked with ideas of right understanding, proper conduct, and spiritual or religious correctness. By the 20th century, the word shifted into a more social and cultural semantic field. Its neighbors include lineage or descent (පෙළපතකින් \ pɛlʌpʌθʌkɪn), Christian (ක්‍රිස්තියානි \ krɪsθɪjɑːnɪ), through action (ක්‍රියාවෙන් \ krɪjɑːvɛn), through language (භාෂාවෙන් \ bʰɑːʃɑːvɛn), and uneducated people (නූගතුන් \ nuːgʌθʊn). In this period, 'සම්යක් \ sʌmjʌk' is used more in relation to social identity, cultural practices, and modes of expression rather than strictly religious or doctrinal correctness.

For the `FastText` embeddings using the neural procrustes alignment approach, the three selected examples are 'රැස් \ ræs', 'සංඛ්‍යාන \ sʌŋkʰyʌɑːθʌ',

Word	Similarity	Ref Century	Ref Century Neighbors	Target Century	Target Century Neighbors
රැස් \ ræs	0.7112	14th	රැස්කරවා \ ræskʌrʌvɑː, පිළිගෙණ \ pɪlɪgɛnʌ, පර්යංකයෙන් \ pʌrjʌŋkʌjɛn, මධ්‍යයෙහි \ mʌðʰyʌjɛhɪ, වේගයෙන් \ veːgʌjɛn	15th	විනාස \ vɪnɑːsʌ, ආයු \ ɑːju, රහස්‍ය \ rʌhʌsyʌ, රත්න \ rʌθnʌ, බුදුරැස් \ buðuræs
සංඛ්‍යාත \ sʌŋkʰyʌɑːθʌ	0.7178	13th	ආකාශ \ ɑːkɑːʒʌ, බෝධිය \ bɔːðʰɪjʌ, අමාත්‍ය \ ʌmɑːθyʌ, විශේෂ \ vɪʒɛʃʌ, කුලෙහි \ kʊlɛhɪ	19th	සංඛ්‍යාවක් \ sʌŋkʰyʌɑːvʌk, ක්‍රමයෙන් \ krʌmʌjɛn, ශරීරයෙන් \ ʒʌriːrʌjɛn, ප්‍රමාණයෙන් \ prʌmɑːnʌjɛn, කාරකයෙහි \ kɑːrʌkʌjɛhɪ
රාජ \ rɑːdʒʌ	0.7335	19th	මංගල \ mʌŋgʌlʌ, ගරුඬ \ gʌrʊdʌ, අතීත \ ʌθiːθʌ, දන් \ ðʌθ, වර්ෂ \ vʌrʃʌ	20th	පුරාතන \ pʊrɑːθʌnʌ, මුඳුන් \ mʊnðʊn, ග්‍රන්ථ \ grʌnθʰʌ, පෙළපතක \ pɛlʌpʌθʌkʌ, නිවටයන් \ nɪvʌtʌjʌn

Table 16: Three lexical items exhibiting semantic drift, analysed using `FastText` embeddings and Neural Procrustes Alignment.

and 'රාජ \ rɑːdʒʌ' as shown in Table 16. The semantic change of the word 'රැස් \ ræs' over the two centuries was discussed in subsection 4.1.

In the 13th century, the word 'සංඛ්‍යාත \ sʌŋkʰyʌɑːθʌ' appears in a semantic environment associated with abstract, spiritual, and social concepts. Its neighbours include sky or space (ආකාශ \ ɑːkɑːʒʌ), enlightenment or awakening (බෝධිය \ bɔːðʰɪjʌ), minister or official (අමාත්‍ය \ ʌmɑːθyʌ), special or distinguished (විශේෂ \ vɪʒɛʃʌ), and clan or lineage (කුලෙහි \ kʊlɛhɪ). In this context, 'සංඛ්‍යාත \ sʌŋkʰyʌɑːθʌ' is linked with ideas of higher status, spiritual significance, and structured social or cosmological order. By the 19th century, the word shifts into a more quantitative and descriptive semantic field. Its neighbors include number (සංඛ්‍යාවක් \ sʌŋkʰyʌɑːvʌk), gradually or in sequence (ක්‍රමයෙන් \ krʌmʌjɛn), from the body (ශරීරයෙන් \ ʒʌriːrʌjɛn), by amount or measure (ප්‍රමාණයෙන් \ prʌmɑːnʌjɛn), and in relation to an agent or doer (කාරකයෙහි \ kɑːrʌkʌjɛhɪ). In this period, 'සංඛ්‍යාත \ sʌŋkʰyʌɑːθʌ' is more associated with numerical expression, measurement, and structured description rather than abstract or spiritual associations.

In the 19th century, the word 'රාජ \ rɑːdʒʌ' appears in a semantic environment associated with royal and ceremonial concepts. Its neighbours include auspicious or ceremonial (මංගල \ mʌŋgʌlʌ), a mythical bird considered the supreme enemy of snakes (ගරුඬ \ gʌrʊdʌ), past or ancient (අතීත \ ʌθiːθʌ), teeth or wisdom-related imagery (දන් \ ðʌθ), and year or time (වර්ෂ \ vʌrʃʌ). In this context, "රාජ \ rɑːdʒʌ" is linked with ideas of kingship, tradition, and symbolic or ritual associations. By the 20th century, the word shifted into a more historical and textual semantic field. Its neighbours include ancient or traditional (පුරාතන \ pʊrɑːθʌnʌ), crown or top/head (මුඳුන් \ mʊnðʊn), book or manuscript (ග්‍රන්ථ \ grʌnθʰʌ), lineage or genealogy (පෙළපතක \ pɛlʌpʌθʌkʌ), and weak or low-status persons (නිවටයන් \ nɪvʌtʌjʌn). In this period, 'රාජ \ rɑːdʒʌ' is more associated with historical identity, textual tradition, and social hierarchy rather than active royal or ceremonial power.

F.2 Bidirectional Semantic Impact Scoring

The aggregate statistics for diachronic semantic drift, computed using `XLM-R` contextual embeddings across 261 consistent lemmas from the 13th, 14th, 15th, 18th, 19th, and 20th centuries, are presented in Table 17. Overall, the results indicate that semantic drift is relatively limited in magnitude. The average global centroid distances range from approximately 0.0019 to 0.0040, and the corresponding average persistent shifts remain consistently close to the original global distances across all time periods analysed. Furthermore, the percentages of drift reduction mostly fall within the range of 15% to 22%, suggesting that the majority of semantic representations maintain substantial continuity over the centuries. This implies that observed semantic changes are primarily influenced by a smaller subset of high-impact contextual usages rather than widespread semantic restructuring.

The most significant semantic divergence is observed in the permutation $18 \rightarrow 14$, which shows the highest average global distance (0.0040) and a persistent shift (0.0033). This indicates that the linguistic contexts represented between these periods have a greater level of contextual and stylistic divergence. More generally, permutations involving the 18th century tend to demonstrate higher global distances and persistent shifts. This could be due to more substantial contextual variation within these permutations, possibly resulting from polysemous usage patterns, domain-specific lexical environments, stylistic variations, or localised semantic changes affecting certain lemma instances.

In contrast, permutations involving the 19th and 20th centuries tend to show lower centroid distances, particularly $19 \rightarrow 20$ and $20 \rightarrow 19$,

Permutation	Avg Global Distance	Avg Persistent Shift	Avg Drift Reduction %	Drifted Instances	Total Instances
$13 \rightarrow 14$	0.0024	0.0020	16.5698	436	2011
$13 \rightarrow 15$	0.0025	0.0021	16.2042	550	2607
$13 \rightarrow 18$	0.0026	0.0022	16.6016	399	1632
$13 \rightarrow 19$	0.0025	0.0021	19.4989	969	4669
$13 \rightarrow 20$	0.0023	0.0019	17.7503	1207	5858
$14 \rightarrow 13$	0.0029	0.0024	20.1966	1036	4847
$14 \rightarrow 15$	0.0035	0.0029	17.6366	558	2600
$14 \rightarrow 18$	0.0034	0.0029	15.9294	414	1652
$14 \rightarrow 19$	0.0033	0.0027	19.9700	998	4668
$14 \rightarrow 20$	0.0036	0.0030	17.3637	1205	5870
$15 \rightarrow 13$	0.0028	0.0023	19.2560	967	4868
$15 \rightarrow 14$	0.0035	0.0030	16.6044	460	2031
$15 \rightarrow 18$	0.0031	0.0026	15.7542	407	1655
$15 \rightarrow 19$	0.0029	0.0024	18.7192	976	4674
$15 \rightarrow 20$	0.0026	0.0022	17.1016	1101	5885
$18 \rightarrow 13$	0.0036	0.0031	17.5780	991	4860
$18 \rightarrow 14$	0.0040	0.0033	16.9774	464	2037
$18 \rightarrow 15$	0.0037	0.0031	17.2689	582	2638
$18 \rightarrow 19$	0.0039	0.0033	17.0748	1024	4710
$18 \rightarrow 20$	0.0036	0.0031	17.1668	1214	5881
$19 \rightarrow 13$	0.0024	0.0020	19.1956	986	4869
$19 \rightarrow 14$	0.0028	0.0024	17.8757	415	2004
$19 \rightarrow 15$	0.0025	0.0021	16.3440	558	2617
$19 \rightarrow 18$	0.0029	0.0024	15.9569	375	1616
$19 \rightarrow 20$	0.0019	0.0016	19.8102	1220	5858
$20 \rightarrow 13$	0.0021	0.0017	21.7573	966	4852
$20 \rightarrow 14$	0.0031	0.0026	17.7576	446	2011
$20 \rightarrow 15$	0.0020	0.0016	18.6148	562	2583
$20 \rightarrow 18$	0.0028	0.0023	15.7908	367	1628
$20 \rightarrow 19$	0.0019	0.0015	19.5775	874	4643

Table 17: Summary Statistics of XLM-R Embeddings for Diachronic Semantic Change of 261 Consistent Lemmas.

both of which produce the smallest average global distances (0.0019). These lower values indicate stronger semantic continuity between modern periods, likely reflecting increased standardisation, reduced orthographic variation, and greater stabilisation of lexical usage in later Sinhala textual traditions.

The highest average drift reduction percentage occurs in the permutation $20 \rightarrow 13$ at 21.7573%, closely followed by $14 \rightarrow 13$ at 20.1966%. These significant reductions indicate that comparisons involving the 13th century are particularly sensitive to isolated high-impact contextual usages. In these instances, a relatively small number of innovative, stylistically distinct, or domain-specific cases can disproportionately affect the overall semantic direction of the lemma representations.

The analysis of drifted instance counts uncovers significant trends at the corpus level. Permutations that include the 19th and 20th centuries consistently show the highest number of drifted instances, often exceeding one thousand occurrences. Despite this, these permutations maintain relatively low average distances and consistent shifts. This suggests that while semantic variation becomes more prevalent in modern corpora due to greater textual coverage and increased lexical diversity, the extent of semantic deviation per instance remains relatively moderate.

The findings indicate that diachronic semantic change in Sinhala, as represented by XLM-R embeddings, is characterised more by localised contextual innovations rather than uniform lexical replacement. These innovations often arise from doctrinal, poetic, or stylistically distinct usages. The small differences between global distances and ongoing shifts across variations suggest that core semantic structures remain largely stable over time. Meanwhile, significant contextual outliers play a major role in the observed semantic drift.

The semantic trajectory of the lemma ඥාන \ nyaːnʌ shows a clear shift from the 20th-century anchor environment to the 15th-century target environment. In the 20th century, the term is used in a broad, general sense of "knowledge," mainly associated with education and learning, as seen in expressions like ශාස්ත්‍ර ඥානය \ ʒasθrʌ nyaːnʌjʌ. In the 15th-century

target data, ඥාන \ nyɑːnʌ is instead embedded in a Buddhist doctrinal framework, where it refers to specific categories of enlightened or supernormal knowledge. This is supported by informers that includes structured lists such as ෂට් අසාධාරණ ඥානය \ ʃʌt ʌsɑːðʰɑːrʌnʌ nyɑːnʌjʌ, චුද්දශ බුද්ධ ඥානය \ tʃʊððʌʒʌ bʊððʰʌ nyɑːnʌjʌ, and ෂට් අභිඥාය \ ʃʌt ʌbʰɪnyɑːjʌ, showing that the term is used within formal religious classifications. The main informer responsible for the large 64.38% structural drift reduction is this dense canonical enumeration context. It restricts the meaning of ඥාන \ nyɑːnʌ to a specialised religious and philosophical sense, rather than the more general educational sense found in the modern period.

The semantic trajectory of the lemma වැසි \ væsɪ shows that the aggregate usage derived from both anchor and target sentences is relatively stable, with the term primarily referring to "rain" in both literal and metaphorical contexts, including extended uses such as 'දහම්වැසි \ ðʌhʌmvæsɪ' where it retains a structured figurative sense within a coherent conceptual frame. However, the informer data introduces two strongly poetic usages, namely "හෙන දියහැලි වැසිය \ hɛnʌ ðɪjʌhælɪ væsɪjʌ" and "මිණි නිල් කැරැල්ලෙන් නැගිරැස් නුබ වැසිය \ mɪnɪ nɪl kærælleːn nægɪræs nʊmbʌ væsɪjʌ", where වැසි \ væsɪ is embedded in highly figurative and imagistic compounds associated with sky, lightning, and ornamental visual imagery. These poetic instances contribute to the observed 45.97% drift reduction by shifting the lemma from a relatively stable literal/metaphorical-religious usage in the aggregate data toward a more condensed poetic register, where meaning is extended through imagery rather than functional or doctrinal usage.

The semantic trajectory of the lemma නාම \ nɑːmʌ shows a relatively stable usage across the aggregate of anchor and target environments, where the term primarily functions in the general sense of "name" or "designation," typically appearing as a grammatical marker for labelling, identification, or formal reference. In both periods, the term is used in a largely referential capacity, as seen in expressions like නාමයෙන් \ nɑːmʌjɛn, where it serves to introduce or specify an identified entity without substantial semantic expansion. In the informer data, නාම \ nɑːmʌ is instead embedded in highly formal and inscriptional registers, where it appears within extended honorific and documentary naming structures. This is supported by contextual windows such as "පූර්ව නාම ප්‍රකාශිත ජයබාහු මහා ස්ථවිරපාදයෝ... \ puːrvʌ nɑːmʌ prvʌkɑːʒɪθʌ dʒʌjʌbɑːhʊ mʌhɑː sθʌvɪrʌpɑːðʌjɔː..." and "ජන්මාන්තරයෙහි... යුක්ත වූ නාමයෙන් ක්‍රැර වූ මේ ජ්වරය \ jʌnmɑːnθʌrʌjɛhɪ... jʊkθʌ vuː nɑːmʌjɛn kræːrʌ vʊ meː", showing that the lemma operates within structured classificatory and titular frameworks rather than simple lexical naming. The main informer responsible for the large 64.23% structural drift reduction is this dense epigraphic and doctrinal naming environment. It constrains නාම \ nɑːmʌ from a general referential "name" function into a formalised mechanism of designation within religious, historical, and institutional discourse, rather than the more general labelling function observed in the aggregate modern usage.